

Review

A Survey on Maximum Ratio Combination: Applications, Evaluation and Future Directions

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Abstract: With the rapid development of wireless communications, the occupation of time and frequency resources becomes more crowded. The exploitation of space resources is necessary and the diversity combining techniques have substantial applications. Diversity combining achieves great diversity gains and improves the ability to combat multipath fading, among which the maximum ratio combining (MRC) performs as the optimal linear combining approach. However, MRC suffers from detrimental factors such as channel fading and no Gaussian noise in practical scenarios. In this paper, we focus on a comprehensive investigation of MRC. Starting from the MRC principle and system model, we summarize typical scenarios and analyze the channel fading statistics. For the influential factors, we further review related literature on channel correlation, cochannel interference (CCI) and impulsive noise. Major performance criteria and performance bounds are derived and compared. MRC confronts new developing challenges and the major development directions are reviewed. The paper finally discusses recent works and open problems for MRC applications and development. Emerging techniques such as artificial intelligence provide novel solutions for MRC performance improvements. The paper aims to present a summarized insight to assist readers in clarifying the analyzed methodology of MRC, so as to motivate new technology integration and extensive applications of advanced communication systems.

Keywords: maximum ratio combining; diversity combining; channel fading; impulsive noise; deep learning



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1. Introduction

Diversity combining techniques are the prominent ways to improve the reliability of communication systems, which are effective to combat multi-path fading and achieve diversity gains from the combinations of dominant signal and replicas based on the fact that asynchronous deep fading occurs [1]. Among diversity combining schemes, maximum ratio combining (MRC) is the optimum to achieve the highest signal-to-noise ratio (SNR) in the presence of Gaussian noise over independent and identically distributed (i.i.d) channels. MRC has been widely involved and analyzed in communication fields such as wireless cellular systems, power line communication (PLC), underwater acoustic communications, and relevant applications of MRC-aided communication systems [2–14]. MRC compensates phase shifts in the complex gains and sums the SNR from each individual subbranch for coherent reception with independent fading.

With different weighted rules, suboptimal linear combining schemes are proposed with lower complexity. Equal gain combining (EGC) equally weights subbranch signals,

where only the phases of the channel are needed [15]. The selection combiner (SC) selects the signal from the diversity branches with the largest instantaneous SNR. Generalized selection combining (GSC) chooses the m largest signals from the total L diversity branches and achieves suboptimal performance with lower complexity [16]. Considering the tradeoff between implementation complexity and improved SNR performance, hybrid selection/MRC (HS/MRC) selects the L strongest signals according to the signal-envelope amplitude from the available N diversity branches and coherently combines them using the MRC method [17]. Sub-branch hybrid selection/MRC (QBHS/MRC) splits the receive branches into in-phase and quadrature subbranches and achieves better performance than traditional HS/MRC [18]. In the Threshold-based hybrid selection/MRC (T-HS/MRC) method [19], the virtual branch technique is introduced to convert ordered-branch variables into i.i.d virtual branches.

From the perspective of diversity resources, the utilization of signals with independent fading is exploited from the time, frequency and space domain. In the time domain, the time duration is separated by many individual time slots and the same information is transmitted through specified time intervals separated by the coherent time. The time diversity is applied on the Rake receiver, and the resolvable multipath components are combined to increase the received SNR [20]. The frequency diversity transmits the same information through different frequency bands separated by the coherence bandwidth at the same time. Frequency diversity multiplexing allows different information to be transmitted in the given frequency spectrum. One of the most representative techniques is orthogonal frequency division multiplexing (OFDM), which is a popular technique to combat multipath fading by separating the bandwidth into parallel subcarriers and improving the tolerance to time delay. Spatial diversity aims to improve the reliability without cost of time and frequency resources and is implemented by utilizing multiple antennas to create uncorrelated propagation paths.

Channel state information (CSI) is required for the maintenance of diversity gains. However, in practical applications, CSI is partially known. Also, non-Gaussian noise and other detrimental influential factors exist. Although there are many technical papers on diversity combining, to the best of our knowledge, there are few comprehensive survey papers for diversity combining especially the MRC and this paper is to fill this gap. We study the leading research on MRC. The profiles of the development of traditional or emerging communication systems with MRC application are presented, which may be non-all-inclusive but representative. The structure of this paper is shown in Figure 1. The major contributions of this paper are summarized as follows:

- The application scenarios of diversity combining in terms of space, time and frequency, i.e., Multiple Input Multiple Output (MIMO), Rake receiver and OFDM are reviewed. Based on these schemes, the system model and mathematical analysis model are abstracted and formulated. In addition, we highlight the spatial diversity and the system models for different scenarios in the presence of Gaussian noise, CCI, and impulsive noise, are presented.
- Performance analysis is critically reviewed and summarized. The commonly used performance metrics and channel fading statistics are introduced, based on which the performance of MRC under different channel fading conditions is summarized and derived. The major influential factors for MRC are analyzed in terms of channel correlation, CCI and impulsive noise.
- We investigate a series of MRC-based or modified MRC techniques. Open issues in emergent communication techniques and the development of MRC-related techniques are outlined. Future research directions for MRC integrated with new techniques such as deep learning are reviewed to help readers clarify the possible applications, various representative scenarios and the prospect for future MRC-aided systems.

The remainder of the paper is organized as follows. Section 2 presents the application framework of MRC in multi-channel communication systems, i.e., MIMO, Rake receiver and OFDM. Section 3 introduces typical MRC application scenarios and analyzes the in-

fluent factors including channel fading statistics, CCI and impulsive noise. In Section 4, the performance metrics and performance analysis for MRC over typical channel fading scenarios are analyzed and summarized. Section 5 analyzes possible development directions and open issues considering the emerging MRC or MRC-aided wireless systems. In Section 6, conclusions are made.

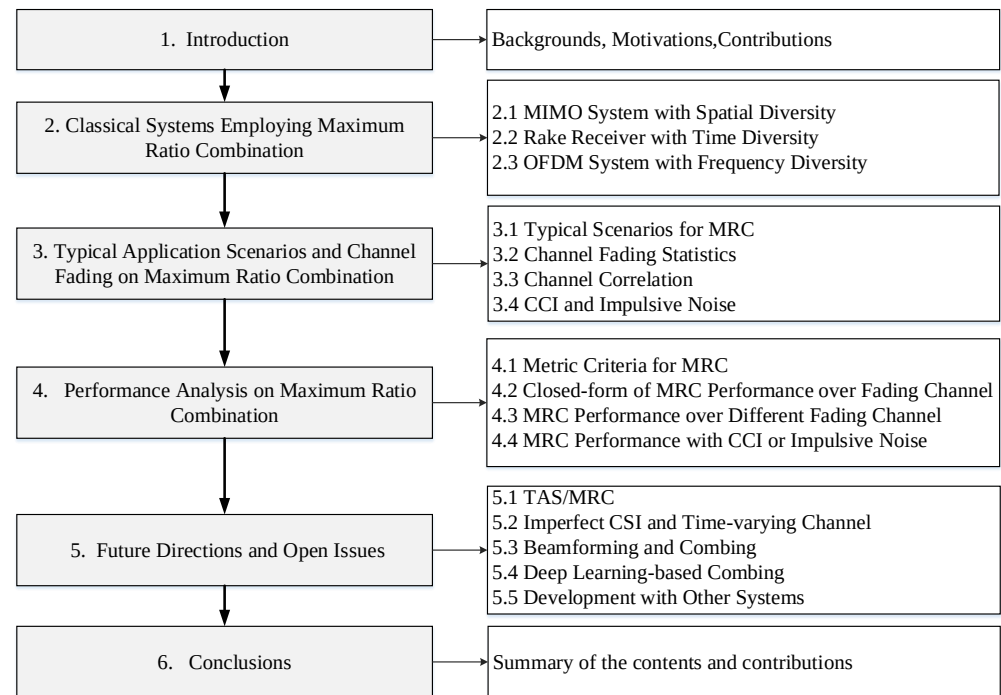


Figure 1. The structure of this paper.

2. Classical Systems Employing Maximum Ratio Combination

Due to the inhomogeneous characteristics of transmission media, multipath components are generated in the angular domain or delay domain, and the resolvable components are combined to improve signal quality. Diversity gain is achieved from different and independent paths, and resolvable paths can be constructed by dividing available space, time and frequency resources. For time diversity, multiple time slots are divided according to the channel coherence time. For frequency diversity, subbands are divided in terms of the channel coherence bandwidth. For spatial diversity, multiple receiver antennas are deployed without the cost of time and frequency resources, and require more computations from the hardware. In this section, we review typical diversity systems employing diversity combining by considering multiple channels such as multiple paths, multiple antennas and multiple frequency subchannels in the light of common communication technologies, and the relevant applications of MRC-aided diversity systems are surveyed.

2.1. MIMO System with Spatial Diversity

MIMO systems are implemented by equipping multiple antennas at the transmitter and receiver. The MIMO system shows high potential in communication reliability and high data rate [21]. A typical MIMO system is shown in Figure 2. In Figure 2 N_t antennas and N_r antennas are equipped at the transmitter and the receiver end, respectively. The data are transmitted through the $N_t \times N_r$ dimensional MIMO channel.

The input-output relation is given by

$$\mathbf{r} = \mathbf{H}\mathbf{s} + \mathbf{n} \quad (1)$$

where the input signal is $\mathbf{s} = [s_1, s_2, \dots, s_{N_t}]^T$ and the received signal $\mathbf{r} = [r_1, r_2, \dots, r_{N_r}]^T$. $\mathbf{n} \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ is the Gaussian noise vector, following the circularly symmetric complex

Gaussian distribution. $\mathbf{H}$ denotes the MIMO channel, and the element H_{ij} represents the subchannel from the i th transmitter to the j th receiver, where $1 \leq i \leq N_t$ and $1 \leq j \leq N_r$. With the assumption of an independent path between the transmitters and receivers, the diversity is scaled up to $N_t \times N_r$. In special cases, the MIMO model degrades to the generalized SIMO when $N_t = 1$ and the MISO system when $N_r = 1$.

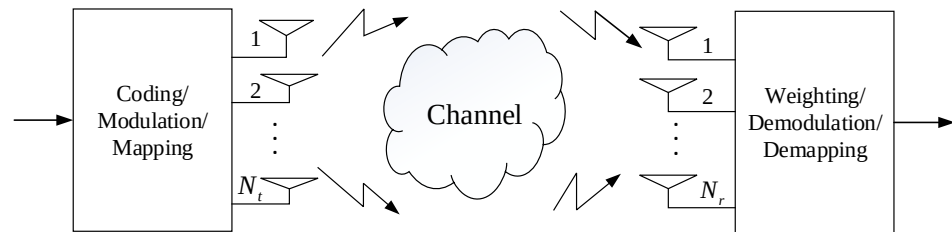


Figure 2. Block Structure of MIMO system.

2.2. Rake Receiver with Time Diversity

Multipath signals occur due to obstacles or irregular channel geometry. At the receiver end, the multipath signals arrive and can be combined efficiently to recover the original signal. Rake receiver was first proposed by Price and Green in 1958 [20]. It resolves the multipath components and uses matched filter as a finger to collect multiple signal replicas to improve the reliability of the received signal. The linear Rake receiver coherently combines the multipath energies using weights corresponding to channel coefficients. When channel parameters are perfectly known, MRC is considered as the optimal combining scheme to maximize the received SNR in the presence of Gaussian noise, which has been employed to improve the performance of code division multiple access (CDMA) system with higher accuracy [22]. The Direct Sequence (DS)-CDMA systems have been widely applied to wireless communication systems based on direct sequence spread spectrum (DSSS) and achieve enhanced privacy, resistance to multipath fading and high spectral efficiency [23].

2.3. OFDM System with Frequency Diversity

OFDM has attracted considerable interest from the research community due to its ability to combat the frequency-selective channel by splitting channel bandwidth into parallel subchannels. OFDM converts the frequency-selective channel into multiple flat-fading subchannels, and the received signal is the sum of all individual received signals from subchannels. The OFDM system is shown in Figure 3. For the frequency-selective channel, OFDM suffers from inter-symbol interference (ISI) and is sensitive to inter-carrier interference (ICI). To mitigate ISI, the guard interval, such as the cyclic prefix is appended ahead of the data frame. For ICI, the work [24] proposed that diversity combining can be employed to reduce the negative effects of ICI.

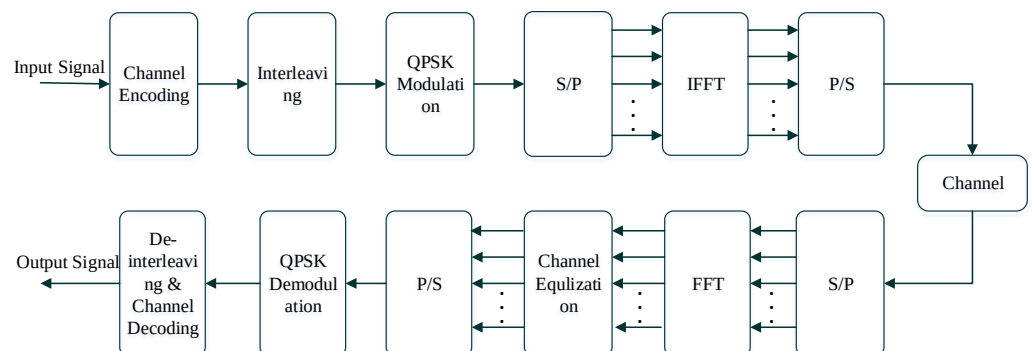


Figure 3. Block Structure of OFDM system.

3. Typical Application Scenarios and Channel Fading on Maximum Ratio Combination

The multichannel signals suffer from multipath fading, external noise and interference. To mitigate the effects of multipath fading and interference, many researchers have studied how to choose the combined weights for performance and capacity improvement. MRC is still a powerful competitor compared with other combining strategies. MRC achieves diversity gains by utilizing the multichannel data according to the linear combining rule. To facilitate the analysis in this paper, the multichannel system model from the spatial diversity system is adopted to illustrate the principle and scenarios of MRC as shown in Figure 4. The model includes a single primary user with one antenna and a receiver equipped with L diversity branches. For the l th subbranch, signal passes through channel with path amplitude $|h_l|$, time delay $|\tau_l|$ and suffers noise n_l . At the receiver end, the combining weight for the l th antenna is $|\alpha_l|e^{j\theta_l}$. In this section, typical scenarios are first presented, i.e., Gaussian noise-limited scenario, CCI-limited scenario as well as impulsive noise-limited scenario. Then, the statistics of multipath fading channels are elaborated.

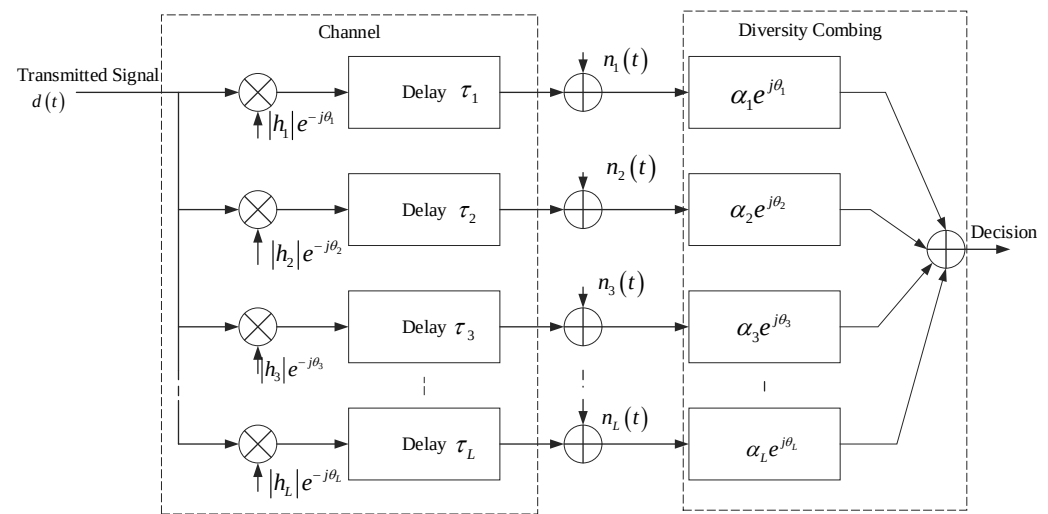


Figure 4. Multichannel system model.

3.1. Typical Scenarios for MRC

The deployment and application of MRC-aided communication may occur under various conditions in real scenarios. In this part, we analyze major factors for MRC analysis. To keep credible performance verification under real-world environments, models are considered that can help give reliable estimations.

3.1.1. Gaussian Noise-Limited Scenario

The signal suffers from channel fading and Gaussian noise. In Gaussian noise-limited conditions, MRC achieves the optimum SNR. The combined received signal is given by

$$\mathbf{r} = \mathbf{h}_0^H \mathbf{y} = \mathbf{h}_0^H \mathbf{h}_0 \mathbf{d}_0 + \mathbf{h}_0^H \mathbf{n}, \quad (2)$$

where $\mathbf{d}_0$ is the transmitted signal, and $\mathbf{h}_0$ is the complex channel subject to i.i.d fading. $\mathbf{n}$ denotes the Gaussian noise vector. For L branches, the received signal $\mathbf{r} = \{\mathbf{r}_l(t)\}_{l=1}^L$, and the instantaneous SNR is

$$\gamma = P_0 \frac{\mathbf{h}_0^H \mathbf{h}_0}{N_0} = \frac{P_0}{N_0} \sum_{i=1}^L |h_0^i|^2, \quad (3)$$

where P_0 represents the signal power and h_0^i is the channel amplitude coefficient of the i th subbranch.

3.1.2. CCI Limited Scenario

Frequency resources are scarce for large-scale networks such as wireless cellular land-mobile radio systems, which require large system capacity. Frequency reuse is an unavoidable method to increase spectrum efficiency at the cost of limited service quality due to CCI resulting from shared spectrum for different users and radiation of signals from the adjacent channels. Received signals include the replicas of the desired signal and the interfering signals. CCIs have more detrimental effects than Gaussian noise. Considering CCI with negligible thermal noise, the scenario is referred to as CCI-limited [25]. In a CCI-limited scenario, the received signal consists of components from both the source signal and interference given by

$$\mathbf{r} = \sqrt{P_0}\mathbf{h}_0\mathbf{d}_0 + \sum_{n=1}^N \sqrt{P_n}\mathbf{h}_n\mathbf{d}_n, \quad (4)$$

where P_0 and P_n ($n = 1, 2, \dots, N$) represent the powers of desired source and interference, respectively. N denotes the number of interfering sources. $\mathbf{d}_0$ and $\mathbf{d}_n$ represent the transmitted data from the desired source and interfering branch. Similarly, $\mathbf{h}_0$ and $\mathbf{h}_n$ represent the complex channel gains experienced by the desired source signal and interfering signal, respectively. The instantaneous signal-to-interference ratio (SIR) is shown as

$$\gamma = \frac{P_0|\mathbf{h}_0^H\mathbf{h}_0|^2}{\sum_{n=1}^N P_n|\mathbf{h}_n^H\mathbf{h}_n|^2}. \quad (5)$$

3.1.3. Gaussian Noise and CCI Scenarios

In the CCI-limited scenario, there is an important assumption that the effect of Gaussian noise is negligible, which is acceptable when the number of receiver antennas is less than or equal to the number of interferences. In practical applications, the received signal is the combination of the desired signal, CCI and Gaussian noise. In the presence of the CCI and Gaussian noise, the system model and the instantaneous signal-to-interference-noise ratio (SINR) are shown as

$$\mathbf{r} = \sqrt{P_0}\mathbf{h}_0\mathbf{d}_0 + \sum_{n=1}^N \sqrt{P_n}\mathbf{h}_n\mathbf{d}_n + \mathbf{n}, \quad (6)$$

and

$$\gamma = \frac{P_0|\mathbf{h}_0^H\mathbf{h}_0|^2}{\sum_{n=1}^N P_n|\mathbf{h}_n^H\mathbf{h}_n|^2 + \sigma^2}, \quad (7)$$

where σ^2 represents the noise power and keeps identical for each subbranch.

3.1.4. Impulsive Noise Scenario

The optimal performance of MRC is based on the assumption of Gaussian noise. However, in practical applications, such as the lightning discharges in PLC, the network interference in wireless communication, the shrimp snapping noise and malicious jamming in underwater acoustic communications [26,27], impulsive noise is required to be considered in the diversity techniques [28,29]. In the presence of impulsive noise, the system model is given by

$$\mathbf{r} = \sqrt{P_0}\mathbf{h}_0\mathbf{d}_0 + \mathbf{i}, \quad (8)$$

where $\mathbf{i}$ represents the impulsive noise vector. In the presence of impulsive noise, the linear combiner will be the sub-optimal receiver, and the diversity gain will be degraded. The conventional design and performance analysis will be overturned with the incorporation of impulsive noise. A new analysis model is required to be resumed and a new performance bound should be given.

3.2. Channel Fading Statistics

In addition to the interference and noise, the signal experiences complicated channel fading including path loss, shadowing (i.e., large-scale variations of the mean power of the signals) and multipath fading. Multipath fading belongs to the small-scale fading. It varies fast and causes short-term signal variations, which results in destructive and constructive combinations of signal replicas from the reflection, scattering, and diffraction. The combined effects make it difficult to depict the channel characteristics. Understanding channel characteristics is a key step in designing and evaluating the communication system, and accurate channel modeling provides a powerful impetus for the design of the communication system. Statistical modeling is considered a powerful tool to characterize the channel effects. In this subsection, the principal characteristics and statistical channel models are reviewed as well as their particular propagation environments and underlying communication scenarios.

3.2.1. Rayleigh Fading

Rayleigh fading can model the fading channel over where large numbers of obstacles exist and have non-line-of-sight (NLOS) signals. The superposition of independent statistical signals presents Rayleigh distribution in terms of the envelope amplitude and the phase follows uniform distribution in the range of $[0, 2\pi]$. The channel amplitude variable of Rayleigh distribution is constructed by the Gaussian constituents following $\mathcal{CN}(0, \sigma^2)$. The probability density function (PDF) of Rayleigh distributed channel coefficient r is given by

$$f(r) = \begin{cases} \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}} & r > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (9)$$

where σ^2 is the variance of Gaussian component.

Rayleigh fading channel is generally used to describe the propagation of reflected and refracted paths through the troposphere, urban environment, ship-to-ship radio link and so on.

3.2.2. Rice Fading

Different from Rayleigh fading, Rice fading can describe the scenario when there is a line-of-sight (LOS) component that is stronger than other scattering components. The rice variable can be formed by two different Gaussian random variables following $\mathcal{CN}(\mu_1, \sigma^2)$ and $\mathcal{CN}(\mu_2, \sigma^2)$ with the independent mean values μ_1, μ_2 , respectively. The PDF of Rice distribution is given by

$$f(r) = \begin{cases} \frac{r}{\sigma^2} I_0\left(\frac{sr}{\sigma^2}\right) e^{-\frac{r^2+s^2}{2\sigma^2}} & r > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (10)$$

where $s = \sqrt{\mu_1^2 + \mu_2^2}$, $I_n(\cdot)$ is the modified Bessel function of the first kind and order n , and $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$ is the Gamma function. The rice factor is defined as $K = \frac{s^2}{2\sigma^2}$, which means the ratio of dominant power s^2 and scattering component power $2\sigma^2$. When $K = 0$, Rice distribution reduces to Rayleigh distribution.

Rice fading is usually used to model the microcellular urban and suburban land-mobile, picocellular indoor, factory environment, the satellite link and ship-to-ship radio link with a dominant LOS path.

3.2.3. Hoyt Fading

Hoyt distribution (also called Nakagami- q) is used to model wireless channels with the in-phase and quadrature signal components which have zero means and unequal variances [30]. Hoyt distribution provides better modeling results in more severe channel

fading conditions for the wireless communication systems [31]. The PDF of Hoyt fading is given by

$$f(r) = \begin{cases} \frac{(1+q^2)r}{q\Omega} e^{-\frac{(1+q^2)^2}{4q^2\Omega}} I_0\left(\frac{(1-q^4)r^2}{4r^2\Omega}\right) & r > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (11)$$

where $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and q denotes the fading parameter in the range of $[0, 1]$. $\Omega = \mathbb{E}[r^2]$ is the average SNR. When $q = 0$, Hoyt fading represents one-sided Gaussian fading, and Rayleigh fading happens when $q = 1$.

Hoyt fading is commonly used to model signal fading due to strong ionospheric scintillation in satellite communications, or fadings which are more severe than Rayleigh with heavy shadowing.

3.2.4. Nakagami- m Fading

Nakagami- m distribution has attracted great attention from researchers due to its expression elegance and versatility and has been verified as a good fitting for empirical methods based on field measurements. Nakagami- m distribution is a two-parameter distribution. The PDF of Nakagami- m fading is given by

$$f(r) = \begin{cases} 2\left(\frac{m}{\Omega}\right)^m \frac{r^{2m-1}}{\Gamma(m)} e^{-\frac{m}{\Omega}r^2} & x > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (12)$$

where the Nakagami fading parameter $m = \mathbb{E}^2[r^2] / \text{var}(r^2)$, the average SNR $\Omega = \mathbb{E}[r^2]$, and $\mathbb{E}(r^v) = \frac{\Gamma(m+v/2)}{\Gamma(m)} \left(\frac{\Omega}{m}\right)^{v/2}$. $\Gamma(m) = \int_0^\infty x^{m-1} \exp(-x) dx$ is the Gamma function.

As a more generalized distribution, Nakagami- m fading is more flexible and more generalized to match the real environmental data than Rayleigh or Rice distribution or Hoyt distribution. By setting the parameter m , Nakagami- m fading can consist of one-side Gaussian distribution ($m = 0.5$), Rayleigh distribution ($m = 1$) and non-fading additive white Gaussian noise (AWGN) channel ($m \rightarrow \infty$). Rayleigh and Hoyt are regarded as the special cases for Nakagami- m fading with $m = 1$ and $m < 1$, respectively. Also, there are relations between the Nakagami- m fading parameter m and Rician factor K . Nakagami- m fading can approximate and generate the corresponding Rician signal with the relation of $m = \frac{(1+K)^2}{2K+1}$ and $\sigma^2 = \frac{\Omega}{2} \left(1 - \sqrt{1 - m^{-1}}\right)$. The relation holds on the condition that $m \geq 1$ and is not suitable for the parameter range $1/2 \leq m < 1$. When $m = 1$, i.e., the corresponding parameter $K = 0$, the Nakagami- m distribution degrades to the Rayleigh distribution. Nakagami fading parameter m is considered as integer-valued. In practice, the fading parameters may not be integer-valued. The Nakagami- m fading channel has been used to describe the statistical characteristics of the signal envelopes for urban and digital mobile radio channels.

3.2.5. $\eta - \mu$ Fading and $\kappa - \mu$ Fading

Most fading distributions are derived under the assumption that a homogeneous diffuse scattering field originates from randomly distributed scattering points. Besides Nakagami fading, more generalized fading distributions namely $\kappa - \mu$ distribution and $\eta - \mu$ distribution were first introduced in [32]. $\eta - \mu$ and $\kappa - \mu$ generalized fadings are fully characterized by physical fading parameters and match well with the experimental data under the proper assignment of parameters. $\eta - \mu$ and $\kappa - \mu$ fadings model the small-scale variations in the channel for the NLOS and LOS conditions, respectively.

The PDF of $\eta - \mu$ distribution is given by

$$f(r) = \frac{4\sqrt{\pi}\mu^{\mu+\frac{1}{2}}h^\mu r^{2\mu}}{\Gamma(\mu)H^{\mu-\frac{1}{2}}\Omega^{\mu+\frac{1}{2}}} \exp\left(\frac{-2\mu hr^2}{\Omega}\right) I_{\mu-\frac{1}{2}}\left(\frac{2\mu Hr^2}{\Omega}\right), \quad (13)$$

where $\mu > 0$ represents the number of multipath clusters and $\mu = \frac{\mathbb{E}^2\{\gamma\}}{2\text{var}\{\gamma\}} \left[1 + \left(\frac{H}{h} \right)^2 \right]$. The model includes two physical models depending on the physical phenomenon referred to as Format 1 and Format 2. The parameter η is the ratio of in-phase power to the quadrature scattered waves in each multipath cluster. In the model of Format 1, $h = (2 + \eta^{-1} + \eta)/4$ and $H = (\eta^{-1} - \eta)/4$. $\eta \in (0, \infty)$ and the in-phase and quadrature-phase components of the fading signal are independent of each other and have different average powers. Format 2 assumes the in-phase and quadrature-phase components in each cluster are correlated with each other and have identical power. $h = 1/(1 - \eta^2)$ and $H = \eta/(1 - \eta^2)$, where $-1 < \eta < 1$ represents the correlation coefficient between in-phase scattered waves and quadrature scattered waves in each multipath cluster. The generalization of $\eta - \mu$ fading can cover Hoyt ($\eta = q^2, \mu = 1/2$), Rayleigh, one-sided Gaussian distribution and Nakagami- m fadings ($\eta \rightarrow 0, \eta \rightarrow \infty, \eta \rightarrow \pm 1$).

$\kappa - \mu$ distribution is suitable for the LOS applications. The PDF of $\kappa - \mu$ distribution is given by

$$f(r) = \frac{2\mu(1+\kappa)^{\frac{\mu+1}{2}} r^\mu}{\kappa^{\frac{\mu-1}{2}} \exp(\mu\kappa) \Omega^{\frac{\mu+1}{2}}} \times \exp\left[-\frac{\mu(1+\kappa)r^2}{\Omega}\right] I_{\mu-1}\left[2\mu\sqrt{\frac{\kappa(1+\kappa)}{\Omega}}r\right], \quad (14)$$

where $\kappa > 0$ denotes the ratio between the power of dominant component and the total power of scattered waves. $\mu > 0$ represents the number of multipath clusters. $I_v(\cdot)$ represents the modified Bessel function of the first kind and v th order. $\mu = \frac{\mathbb{E}^2\{\gamma\}}{2\text{var}\{\gamma\}} \cdot \frac{1+2\kappa}{(1+\kappa)^2}$. The generalization of $\kappa - \mu$ distribution covers Nakagami- m ($\kappa \rightarrow 0$ and $\mu = m$), Rayleigh ($\mu = 1$ and $\kappa \rightarrow 0$) and one-sided Gaussian distribution ($\mu = 0.5$ and $\kappa \rightarrow 0$).

To moderately fit the multipath fading in the non-homogeneous scenarios and cancel the approximation of homogeneous diffuse scattering field, $\alpha - \mu$ fading was proposed to explore the linearity of the propagation medium [32]. Within one cluster, the phases of scattered waves are random and have similar delay time spread. It is a rewritten form of the Stacy (generalized Gamma) distribution and is also a generalized distribution similar to the $\eta - \mu$ and $\kappa - \mu$ fading. It includes Gamma, Nakagami- m , exponential, Weibull, one-side Gaussian, and Rayleigh fading. Furthermore, $\alpha - \eta - \mu$ and $\alpha - \kappa - \mu$ distributions [33] represent the small-scale fading model, which describes the multipath fading in the nonlinear medium under NLOS and LOS conditions, respectively. The new generalized shadowed fading model has the analytically statistical form, which can cover $\kappa - \mu$, $\eta - \mu$, Nakagami- m , Rice, Rician shadowed and one-side Gaussian fading models.

3.2.6. Log-Normal Fading

The shadowing effects are commonly referred to as long-term fading, which are generally non-negligible, especially for conditions with dense obstacles in terrestrial mobile communication and land-mobile satellite communication systems. The shadowing effects reflect signal variation around the mean signal levels. The log-normal distribution has been verified as an efficient statistical model to fit the shadow fading from empirical measurements for outdoor and indoor scenarios.

The relationship between the log-normal distribution and the normal distribution is built through the log transformation, i.e., $Y = \ln r$, and Y follows the normal distribution with mean m and variance σ^2 . The PDF of the log-normal distributed envelope is given by

$$f(r) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}r} e^{-\frac{(\ln r - m)^2}{2\sigma^2}} & r \geq 0 \\ 0 & \text{otherwise} \end{cases}. \quad (15)$$

3.2.7. Composite Fading Model

In the process of signal propagation, detrimental effects such as path loss, multipath-fading and shadowing are included. Typical composite fading scenarios exist such as the

wireless cellular communication in the downtown areas with moving pedestrians and vehicles, the land satellite communication subject to urban shadowing, and the underwater acoustic communications with effects from the complex and changeable sea environment including waves, internal turbulence, surface and reflections and refraction. Taking the typical mobile radio communication scenarios as examples, signal propagation suffers from fast fluctuations around the mean signal level and slow variations in the mean signal level itself. For the receivers that cannot average the small-scale fading variations from the multipath fading, the superimposed effects with the shadowed fading are necessary to be taken into account. The composite multipath/shadowing fading distributions are suggested for channel modeling in these scenarios, and many researchers have studied comprehensively for these distributions and brought out many composite fading distributions.

Combined with log-normal shadowing, Rayleigh-lognormal, Rice-lognormal and Nakagami- m -lognormal fading channels were introduced to model the multipath fading distribution superimposed on the lognormal shadowing distribution [34]. Similarly, the Suzuki distribution was proposed as a composite Rayleigh-lognormal distribution to characterize the mobile channel by modeling the root mean square (RMS) value of the fading amplitude as the lognormal distribution. There are many works on the Weibull-lognormal especially for the mobile radio system in the 800/900 MHz frequency range [35], Hoyt-lognormal distribution [36], $\alpha - \eta - \mu$ /lognormal distribution, $\alpha - \kappa - \mu$ /lognormal distribution [37] and so on.

The representation of the log-normal distribution-based composite distributions are intractable mathematically to handle and have complicated representations due to the complicated integral form for performance analysis. Actually, the log-normal distribution can be approximated by the Gamma distribution, which has been an alternative to simplify the representation of the log-normal distribution and accurately model the fading shadowing channel. The Gamma distribution provides better ways to model the composite multipath fading/shadowing model. Many Gamma distribution-based fading models have been derived such as Weibull/gamma composite distribution [38], and $\eta - \mu/\gamma$, $\alpha - \mu/\gamma$ sequentially [39]. The Gamma distribution-based fading models, i.e., the K -distribution and Generalized- K (K_G), have been widely used in radar communication, and model diverse scattering phenomena such as tropospheric propagation of radio waves, the extended tail of radar clutter and optical scintillation from the atmosphere [40,41]. Recently, the two distributions have been studied in the communication field such as PLC, radio wireless communication [42], underwater acoustic communication systems [43] and optical wireless communication. K distribution is equivalent to the mixture of Rayleigh and Gamma fading distributions. Due to the complicated mathematical form of Rayleigh-lognormal distribution, K distribution can be a promising form approximating well with the Rayleigh-lognormal distribution in terms of PDF and mathematical tractability [44]. K_G distribution is more generic and versatile than K distribution. K_G distribution can model a variety of multipath fading and shadowing fading with different model parameters. As a lognormal-based composite distribution, K_G distribution has been verified as a more tractable model compared with the Nakagami- m -lognormal and Rayleigh-lognormal fading [45]. The fading envelope amplitude r following K_G distribution is given by

$$f(r) = \frac{4m^{(\beta+1)/2}x^\beta}{\Gamma(m)\Gamma(k)\Omega^{(\beta+1)/2}}K_\alpha\left[2\left(\frac{m}{\Omega}\right)^{1/2}x\right], r \geq 0, \quad (16)$$

where k and m are the distribution shaping parameters, and $\alpha = k - m$, $\beta = k + m - 1$. $K_\alpha(\cdot)$ is the modified Bessel function of order α . $\Omega = \mathbb{E}\{r^2\}/k$ is the mean power. When using different combinations of k and m , different distributions are fitted. When $k \rightarrow \infty$, it can approximate Nakagami- m distribution. When $m \rightarrow \infty$ and $k \rightarrow \infty$, the no-fading channel is approached [46].

As a special case, when $m = 1$ the K_G distribution reduces to K -distribution, and the resulting PDF is given by

$$f_k(x) = \frac{4}{\sqrt{\alpha}\Gamma(v)} \left(\frac{x}{\sqrt{\alpha}}\right)^v K_{v-1}\left(\frac{2x}{\sqrt{\alpha}}\right), \quad (17)$$

where v is the shape parameter, α is the scale parameter, $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$ is the Gamma function, and K_{v-1} is the modified Bessel function of the second kind of order $v - 1$. $\alpha v = \Omega$. Rayleigh distribution is a special case of K distribution when $v \rightarrow \infty$.

Gamma-Gamma distribution is commonly used to model a mathematically tractable composite fading model. The FSO link suffers from in-homogeneity in temperature, atmospheric pressure, and atmospheric turbulence-induced fading [47]. Among these factors, the turbulence-induced fading, i.e., the fast fluctuations of the irradiance of the optical signals greatly limit the system performance. The Gamma-Gamma distribution is applied on accurate channel modeling to describe the signal distortions by atmospheric turbulence [48]. For radio wireless communication, Gamma-Gamma distribution has been used to model the power statistics in composite fading channels. Gamma-Gamma distribution is equivalent to the squared Generalized- K distribution [49,50] and covers squared K distribution and double-Rayleigh distribution. As the name refers, Gamma-Gamma distribution can be derived by the product of two independent Gamma random variables. The PDF of Gamma-Gamma distribution is given by

$$f(r) = \frac{2(km)^{\frac{k+m}{2}} \gamma^{\frac{k+m}{2}-1}}{\Gamma(m)\Gamma(k)\Omega^{\frac{k+m}{2}}} K_{k-m} \left[2 \left(\frac{km}{\Omega} \gamma \right)^{1/2} \right], \quad (18)$$

where $k \geq 0$ and $m \geq 0$ are the shaping parameters, $\mathbb{E}(r) = \Omega$ is the mean of variable. With different shaping parameters, the Gamma-Gamma distribution approximates other well-known Gamma-based distributions. For instance, when $k \rightarrow \infty$, the Gamma-Gamma distribution reduces to the Gamma distribution. When $m = 1$, it matches the squared K -distributed random variable with PDF given by

$$f(r) = \frac{2k^{\frac{k+1}{2}} \gamma^{\frac{k-1}{2}}}{\Gamma(k)\Omega^{\frac{k+1}{2}}} K_{k-1} \left[2 \left(\frac{k}{\Omega} \gamma \right)^{1/2} \right]. \quad (19)$$

Besides the composite fading above, other composite fading channels are proposed. Rice shadowed fading model is a combination of Rice fading and Nakagami- m shadowing, which has been used to model satellite communication and UAC links. $\kappa - \mu$ shadowed fading model is a model to represent Nakagami- m shadowing and $\kappa - \mu$ fading [51]. These new generalized shadowed fading models have a traceable analytical statistical form that can cover the $\kappa - \mu$, $\eta - \mu$, Nakagami- m , Rice, Rice shadowed and the one-side Gaussian fading models [52].

3.2.8. Fluctuating Two-Ray Fading Model

Due to spectral congestion, wireless communications have developed into 5G and new application scenarios have been envisioned in the 6G era. Millimeter-wave (mmWave) and sub-THz/THz bands provide solutions for high data traffic. Conventional fading models fail to fit the random fluctuations in mmWave and THz bands [53]. The Fluctuating Two-ray (FTR) Fading model was proposed for the channel modeling in emerging mmWave communications, which consists of two fluctuating specular components with random phases plus a diffuse component [54]. This model provides flexibility to describe the randomness of signal fluctuations, which better characterizes the amplitude fading of signal through multiple scatters. FTR fading model has better fitting performance for mmWave communications and can be generalized as various conventional fading distributions by setting different parameters, such as one-sided Gaussian, Rayleigh, Nakagami- m , Hoyt, Rice shadowed, and two-wave with diffuse power (TWDP).

The complex baseband FTR fading channel is expressed as

$$V_r = \sqrt{\zeta} V_1 \exp(j\phi_1) + \sqrt{\zeta} V_2 \exp(j\phi_2) + X + jY \quad (20)$$

where ζ denotes a Gamma-distributed random variable with unit mean. V_1 and V_2 are specular components with constant amplitudes and are modulated by a Nakagami- m random variable. ϕ_1 and ϕ_2 denote uniformly distributed random phases. $X + jY$ represents the diffuse component, where X and Y follow Gaussian distribution with zero mean and variance σ^2 . The PDF of squared FTR random variable ζ is given by [53,55]

$$f(\zeta) = \frac{m^m}{\Gamma(m)} \sum_{j=0}^{\infty} \frac{K^j d_j \zeta_l^j}{(2\sigma^2)^{j+1} \Gamma(j+1)} \exp\left(-\frac{\zeta}{2\sigma^2}\right) \quad (21)$$

and

$$d_j = \sum_{k=0}^j \binom{j}{k} \left(\frac{\Delta}{2}\right)^k \sum_{v=0}^k \binom{k}{v} \Gamma(j-k+m+2v) e^{\frac{\pi(2v-k)i}{2}} \times \left((K+m)^2 - (K\Delta)^2\right)^{-\frac{(j+m)}{2}} P_{j+m-1}^{k-2v} \left(\frac{m+k}{\sqrt{(m+K)^2 - (K\Delta)^2}}\right) \quad (22)$$

where m denotes the shadowing parameter, σ^2 is the variance of diffused components, and $K > 0$ is the ratio of the average power of the dominant component to the power of the diffuse multipath. $\Delta \in [0, 1]$ denotes the relation between the powers of two dominant waves. $P(\cdot)$ denotes Legendre functions of the first kind.

The FTR fading channel provides better fitting results for small-scale fading in mmWave communications than conventional stochastic channel models. As a new-type model, research on the statistical characteristics of FTR fading began in the spring of 2018. The paper [55] derived the PDF and cumulative distribution function (CDF) for the FTR fading. Subsequently, the authors study the statistical characteristics of the sum of squared FTR random variables, which can be applied to the MRC receivers over FTR fading channels [56]. Further, the work [53] provided exact and tractable expressions for the PDF and CDF of the sum of squared FTR random variables, which provides a closed-form expression and better fitting results. The work [57] presented the FTR model as a countable mixture of Gamma distribution and derived new statistical characteristics of the FTR fading model assuming any arbitrary positive values of m . Along with the development of 6G and the understanding of the propagation characteristics deepening, more accurate FTR fading models and analyses keep developing.

3.3. Channel Correlation

Diversity gain depends on the independence and freedom degree. Channel fading is typically assumed as independent when antennas are well separated. The assumption of independent fading presents a coarse evaluation criteria for system design or performance analysis [58]. As the development of the large-scale antenna array for wireless systems, the independence of inter-antennas is destroyed and invalid due to the close-spaced antenna elements.

The correlation among antennas is non-negligible and diversity gain is violated. The effects of correlation cannot be ignored, and the assumption of statistical independence will degrade the diversity gain [59]. In practical applications, the performance of diversity combining suffers from the severity of fading and the correlation type, which requires the correction based on existing results or re-evaluating new influential factors.

According to (1), we assume channel matrix $\mathbf{H}$ is spatially correlated Rayleigh fading and can be decomposed according to the Kronecker correlation structure given by

$$\mathbf{H} = \mathbf{R}^{\frac{1}{2}} \mathbf{H}_w \mathbf{S}^{\frac{1}{2}} \quad (23)$$

where $\mathbf{R} > 0$ and $\mathbf{S} > 0$ denote the receive and transmit spatial correlation matrices, respectively. $\mathbf{H}_w \sim \mathcal{CN}_{N_r, N_t}(\mathbf{0}_{N_r \times N_t}, \mathbf{I}_{N_r} \otimes \mathbf{I}_{N_t})$. To quantify the correlations between the fading signals from each of the diversity branches, the signal parameter ρ is defined to characterize the correlation.

One correlation model is the exponential distribution, where the correlation decreases as the distance between antennas increases. For the exponential correlation model, the (i, j) th entries of $\mathbf{R}$ and $\mathbf{S}$ are given by

$$[\mathbf{R}]_{i,j} = \rho_R^{|i-j|}, [\mathbf{S}]_{i,j} = \rho_S^{|i-j|} \quad (24)$$

where ρ_R and ρ_S represent the correlation coefficients between the two corresponding adjacent antenna elements. The exponential correlation model is common and accurate in wireless communication systems [60–62]. The work [63] has derived the detection performance for exponentially correlated and constant L branches over the Nakagami- m channel for MRC diversity.

The constant correlation model is applied in scenarios where antennas are closely spaced and roughly approximates other correlation models. For the constant correlation model, the correlation coefficients are defined as

$$[\mathbf{R}]_{i,j} = \begin{cases} 1 & i = j \\ \rho_R & i \neq j \end{cases}, [\mathbf{S}]_{i,j} = \begin{cases} 1 & i = j \\ \rho_S & i \neq j \end{cases} \quad (25)$$

where ρ_R and ρ_S reflect average values of the receive and transmit correlation coefficients, respectively, of all off-diagonal entries [64].

The accurate generation of multiple correlated fading channels is important for simulation, verification, evaluation and design for correlated channel diversity systems. With the definitions of correlation coefficients, different fading channels are generated besides the Rayleigh fading in (23), and performance over various fading channels is derived. In the works [65–68], the correlated Rayleigh, the correlated Rician, and the correlated Nakagami- m are studied.

The temporal and spatial correlations of multiple channels in diversity combining and MIMO systems are traditionally described by autocorrelation function (ACF) and cross-correlation functions (CCFs) [69–71]. The work [72] proposed a unified approach to generate multiple envelope processes utilizing given ACFs, CCFs, and marginal PDFs. The Gaussian vector autoregressive (AR) process generates samples and sends them into the inverse transform sampling to output the fading envelope vectors with the given PDF. The correlated Rayleigh fading channels, originating from spatial uncorrelated Rayleigh fading MIMO channel, may be nonzero-mean uncorrelated, zero-mean separately correlated at the receiver and the transmitter, or nonzero-mean separately correlated at both ends. The works [61,73] proposed a separately correlated model that captures the level of spatial correlations in a single parameter. As a different channel model, the work [74] validated the separately correlated channel model, which underestimated the mutual information, while another channel model overestimates the mutual information [75]. Analytical fading channel models can be combined with physical wave propagation-based models, i.e., correlation-based models, such as the Kronecker model [74] and the Weichselberger model [76].

Realistic radio channel models are crucial for performance validation. Most channel models need high computation and simulations, especially for MIMO channel modeling. With the increasing requirement for reliable evaluations, many practical channel simulators appear. To generate accurate radio propagation conditions, the low-cost hardware-based simulators on field-programmable gate array (FPGA) are implemented [77,78]. The work [79] proposed a MIMO-basedband fading channel simulator on FPGA.

3.4. CCI and Impulsive Noise

CCI and impulsive noise belong to external unintentional interference. Both are detrimental to system performance. This part mainly analyzes related research on MRC applications in the presence of CCI and impulsive noise.

Due to finite spectrum resources and high requirements for data rate in cellular systems, frequency reuse introduces CCI. In the same frequency band, the received signals

consist of the desired signal and its replicas, as well as interfering signals radiated from different users located in other positions. In practical applications, CCI can not be ignored and MRC achieves sub-optimal performance. CCI from multiple users dominates system performance and limits the system's capacity. MRC cannot combat the multipath fading and CCI simultaneously. Aiming to solve the issue introduced by CCI, beamforming is an effective technique to suppress CCI by changing the transmit and receive beam patterns. diversity combining technique and adaptive array processing techniques can be integrated to alleviate the interference signals.

Adaptive array beamformer places its nulls to the directions of sources of interferences and mainlobe in the direction of the desired signal. The nulling effects also depend on the number of interferences. The performance of interference cancellation (IC) is constrained by the number of dominant interferences. The beamformer's suppression function is limited especially when the number of interferences exceeds the number of antenna elements. In cellular wireless networks, the number of CCIs is often larger than the number of subbranches, and the adaptive array processing cannot deal with all the interference. Novel IC methods are required to be explored.

Generally, background Gaussian noise is assumed following the central limit theory. However, in the applicable background, impulsive noise needs to be considered. To accurately analyze performance and achieve better performance, impulsive noise models and receiver designs have become research hotspots to combat hostile environments. Popular non-Gaussian noise models can be categorized as Middleton's Class A and Class B model [80], generalized Gaussian model [81], symmetric α -stable (S α S) model as well as the approximated distribution Gaussian mixture (GM) model (also referred as the generalized version of truncated Middleton's Class A noise) [82].

- Middleton's Class-A noise

Middleton's Class-A model is one of the most accurate models of narrowband impulsive noise, which has been widely applied in the wireless communication field for performance analysis and system design. The impulsive noise can be seen as the superposition of the impulsive component and i.i.d the Gaussian component. The occurrences of impulsive components follow Poisson distribution $p(M = m) = \frac{e^{-A} A^m}{m!}$ when the number of independent sources k is large enough. A is the impulsive index that describes the average number of impulses. The PDF of the complex-valued Class-A noise is given by

$$\begin{aligned} p(w) &= \sum_{m=0}^{\infty} p(M = m) p_{x|M=m}(x | k) \\ &= \sum_{m=0}^{\infty} \frac{\alpha_m}{\pi \sigma_m^2} \exp\left(-\frac{|w|^2}{\sigma_m^2}\right), \end{aligned} \quad (26)$$

where $\alpha_m = \exp(-A) A^m / m!$, $\sigma_m^2 = \sigma^2 (m/A + T) / (1 + T)$ and $\sigma^2 = \text{var}(w)$ is the power of noise components. T is the power ratio of Gaussian noise to impulsive noise component. When A is a large value, the impulsive components approximate Gaussian features. When A is smaller, the more impulsive the noise presents. The infinite summation in (26) is totally decided by three parameters σ^2 , T , and A . For the joint vector $\mathbf{w} = [w_1, w_2, \dots, w_L]$, its joint distribution of w_l is the product of the PDF of each component $p(\mathbf{w}) = \prod_{l=1}^L p(w_l)$ given by

$$p(\mathbf{w}) = \sum_{m=0}^{\infty} \frac{\alpha_m}{\pi^L \sigma_m^{2L}} \exp\left(-\frac{\sum_{l=1}^L |w_l|^2}{\sigma_m^2}\right), \quad (27)$$

- S α S noise

α -stable distribution is important in modeling the multiple access interferences in ad-hoc networks, new-field interference in wireless transceivers and impulsive noise in underwater acoustic communications [83–85]. The α -stable noise has no closed-form expression, and its characteristic function is given by

$$\varphi(t) = \exp\{j\mu t - |\sigma t|^\alpha (1 + j\beta \text{sign}(t)w(t, \alpha))\}, \quad (28)$$

$$w(t, \alpha) = \begin{cases} \tan\left(\frac{\pi\alpha}{2}\right) & \alpha \neq 1 \\ \frac{2}{\pi} \log|t| & \alpha = 1 \end{cases}, \quad (29)$$

where $\alpha \in (0, 2]$ is the characteristic exponent, $\beta \in [-1, 1]$ is the skew parameter, $\sigma \in (0, \infty)$ is the scale parameter and $\mu \in (-\infty, \infty)$ is the shift parameter. When $\beta = 0$ and $\mu = 0$, the α -stable distribution is symmetric about μ , which is referred to as the S α S distribution. Then its characteristics function is simplified as

$$\varphi(t) = \exp\{-\sigma|t|^\alpha\}. \quad (30)$$

With the characteristic function, the PDF of S α S is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(t) e^{jtx} dt. \quad (31)$$

- Gaussian-mixture noise

Heavy-tail distribution can be approximated by the GM model. The GM model can be shown as

$$f(w) = \sum_{m=1}^M p_m f_m(w), \quad (32)$$

where M is the number of Gaussian components, and $f_m(w)$ is the PDF of the m th Gaussian component. p_m is the weighted coefficient. $0 \leq p_m \leq 1$ and $\sum_{m=1}^M p_m = 1$. The PDF of the m th Gaussian component is given by

$$f_m(w) = \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left(-\frac{(x - \mu_m)^2}{2\sigma_m^2}\right), \quad (33)$$

where μ_m and σ_m^2 denote the mean and variance of the m th component. When $M = 2$, the model is the two-component GM (GM2) model. When $\sigma_2^2 \gg \sigma_1^2$ and $p_2 < p_1$, the noise samples with larger variance have a lower occurrence ratio, which are seen as the outliers of the background Gaussian noise.

The conventional receiver performs worse in the presence of impulsive noise. To combat the effects of impulsive noise, many optimal or suboptimal designs for receivers are studied [86,87].

4. Performance Analysis on Maximum Ratio Combination

It is necessary to explore the analytical results of performance for practical communication system design over fading channels. MRC tries to maximize the array output SNR and achieves the optimum performance in a noise-limited scenario, and the suboptimal performance in the presence of the CCI. The performance analysis for MRC provides the bound for other diversity combiners. With accurate channel modeling, the performance metrics for MRC have been studied. This section presents major metrics and provides analytical results over different fading scenarios. Moment-generating function (MGF) and PDF-based mathematical tools are always used to obtain the closed-form expression of bit error rate (BER), outage probability and ergodic capacity.

4.1. Metric Criteria for MRC

To measure the effects of channel fading and external noise, performance metrics such as average SNR (ASNR), average bit error rate (ABER), and outage probability are the most common metrics to reveal the nature of communication system behavior.

4.1.1. Average SNR

SNR decides the performance of data detection in a communication system. For the receiver, the channel fading effects are measured appropriately by ASNR. By incorporating the fading statistics of channel fading, the ASNR is defined as [88]

$$\bar{\gamma} = \int_0^{\infty} \gamma f(\gamma) d\gamma, \quad (34)$$

where γ denotes the instantaneous SNR and $f(\gamma)$ is the PDF of γ .

4.1.2. Average Bit Error Rate

The ABER is almost the most common metric for the system performance evaluation especially for different modulation schemes and is defined as the average of the conditional BER over the channel fading variations. ABER is calculated by finite-range integrals with integral related to conditional BER over fading statistics given by

$$p_e = \int_0^{\infty} A Q(\sqrt{B\gamma}) f(\gamma) d\gamma, \quad (35)$$

where $f(\gamma)$ is the PDF of the instantaneous SNR of MRC output. A and B denote different modulation schemes. However, the integral in ABER is difficult to achieve in the closed form due to the nonlinear function like Q -function. $AQ(\sqrt{B\gamma})$ represents the nonlinear function of instantaneous BER conditioned on γ in the AWGN channel according to the modulation and detection schemes.

4.1.3. Outage Probability

Outage probability is another performance metric to measure the transmission quality of a diversity system. Outage probability is defined as the failure rate when the SNR or SINR γ falls below the specified threshold γ_{th} . Outage probability is one of the major considerations in the design of cellular radio systems, and the mathematical presentation is given by

$$p(\gamma < \gamma_{th}) = \int_0^{\gamma_{th}} f(\gamma) d\gamma. \quad (36)$$

4.1.4. Average Channel Capacity

Average channel capacity (ACC) is an important metric, which provides an upper bound for the maximum system capacity. The formula is given by

$$C = \int_0^{\infty} \log_2(1 + \gamma) f(\gamma) d\gamma. \quad (37)$$

4.1.5. Amount of Fading

ABER, outage probability, and ASNR are the most important performance metrics to reveal the behavior of digital communication systems in the presence of fading. To fully capture the diversity gains, the high moments of the output SNR for the diversity combiner exploit the signal envelope fluctuations introduced by fading. Hence, the measured metric, i.e., the amount of fading (AF) is introduced to present the first and second moments of instantaneous SNR. The AF for total output SNR is defined by

$$AF = \frac{\text{var}[\gamma]}{\mathbb{E}[\gamma]^2}. \quad (38)$$

4.1.6. Average Outage Duration

Besides the outage probability, the average outage duration (AOD) is used to describe the average time duration (in seconds) in the state of the outage. AOD provides an alternative view to declare the effect of output SNR, and it is defined by

$$\text{AOD} = \frac{P_{out}}{N(\gamma_{th})}, \quad (39)$$

where $N(\gamma_{th})$ denotes the frequency of outage or the average level crossing rate (LCR) of the output SNR γ at level γ_{th} .

4.1.7. Moment Generating Function-Based Method

For performance metrics, the closed-form expressions are required. For approximated fading distribution, more simple and better distributions are needed to replace the complicated integral. Utilizing the PDF of average SNR is feasible for the i.i.d fading. However, it is difficult to derive the relevant metrics especially when considering more complex fading channels. The MGF-based method has been a unified method for performance analysis due to elegant mathematical representation given by

$$M_{\gamma}(s) = \int_0^{\infty} \exp(-s\gamma) f(\gamma) d\gamma. \quad (40)$$

With the aid of differentiation to MGF, the average SNR is transformed as

$$\bar{\gamma} = \left. \frac{dM_{\gamma}(s)}{ds} \right|_{s=0}. \quad (41)$$

For the MRC analysis, the summations of SNRs from L branches can be replaced by the product of MGFs for subchannels, i.e., $M_{\gamma}(s) = \prod_{l=1}^L M_{\gamma_l}(s)$. Consequently, the closed-form is easily achieved.

For the outage probability, $f(\gamma) = dF(\gamma)/d\gamma$ can be realized by the Laplace transform and replaced by the inverse Laplace transform of $M_{\gamma}(-s)/s$, where $F(\gamma)$ is the CDF. Then the outage probability is evaluated as

$$P_{out} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} \frac{M_{\gamma}(-s)}{s} e^{s\gamma_{th}} ds. \quad (42)$$

For the calculation of ABER, it is difficult to achieve the conditional BER due to the nonlinearity, especially for the multichannel case. The MGF-based method has been used to simplify the analysis and unify ABER analysis for various fading channels.

Besides MGF, powerful mathematical tools are necessary for mathematical results [89,90], such as incomplete generalized MGF, generalized Q function, confluent hypergeometric function ${}_1F_1(\cdot)$ and bivariate confluent hypergeometric function $\Phi_2(\cdot)$, Lauricella multivariate hypergeometric function, general Marcum-Q function, incomplete gamma function, Nuttall-Q function, which can be easily implemented in most popular computing software, i.e., Maple, and Matlab. The Bessel function can further reduce the difficulty and increase the accuracy without truncating the infinite series form. Fox's H-function builds the connection with Meijer's G function. Gaussian hypergeometric functions provide the results in terms of infinite series form but can converge rapidly and steadily and require very few terms for accuracy. Performance analysis can be implemented using different mathematical methods.

4.2. Closed-Form of MRC Performance over Fading Channel

Having covered the typical fading channels and performance metrics, we derive the related PDFs over i.i.d Rayleigh, Rice and K distribution fading channel as shown in Table 1, which summarizes and deduces the formulas of PDF, instantaneous SNR and average SNR over typical statistical channel models.

Performance analysis is an important topic related to MRC, and a large number of papers have studied the topic. Besides channel fading and AWGN, MRC is also constrained by factors such as channel correlation, external impulsive noise and CCI with various powers or correlations. With the coverage of statistical characterizations of the channel

fading model and the performance criterion such as ABER, outage probability, and ACC, we review the related works and derive the performance over different fading channels of MRC as shown in Tables 2 and 3 following the derivation formulas in Table 1.

Table 1. PDF of Amplitude, Instantaneous SNR and Average SNR.

Channel Fading	PDF of Amplitude r , Instantaneous SNR ζ and Average SNR γ
Rayleigh	$f(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}$ $f(\zeta) = \frac{1}{2\sigma^2} e^{-\frac{\zeta}{2\sigma^2}}, \zeta > 0$ $f(\gamma) = \frac{\gamma^{L-1}}{2^L \Gamma(L) \sigma^{2L}} e^{-\frac{\gamma}{2\sigma^2}}$
Rice	$f(r) = \frac{r}{\sigma^2} e^{-\frac{r^2 + s^2}{2\sigma^2}} I_0\left(\frac{sr}{\sigma^2}\right)$ $f(\zeta) = \frac{1}{2\sigma^2} e^{-\frac{s^2 + \zeta}{2\sigma^2}} I_0\left(\frac{s}{\sigma^2} \sqrt{\zeta}\right), \zeta > 0$ $f(\gamma) = \frac{1}{2\sigma^2} \left(\frac{\gamma}{s^2}\right)^{\frac{L-1}{2}} e^{-\frac{s^2 + \gamma}{2\sigma^2}} I_{L-1}\left(\frac{s}{\sigma^2} \sqrt{\gamma}\right), \gamma > 0, K = \frac{s^2}{2\sigma^2}$
K-fading	$f(r) = \frac{4m^{\frac{v+m}{2}} r^{v+m-1}}{\Gamma(m)\Gamma(v)\Omega^{\frac{v+m}{2}}} K_{v-m}\left[2\left(\frac{m}{\Omega}\right)^{1/2} r\right], r > 0$ $f(\zeta) = \frac{2m^{\frac{v+m}{2}} \zeta^{\frac{v+m}{2}-1}}{\Gamma(m)\Gamma(v)\Omega^{\frac{v+m}{2}}} K_{v-m}\left[2\left(\frac{m}{\Omega}\right)^{1/2} \sqrt{\zeta}\right], \zeta > 0$ $f(\gamma) = \frac{2m'^{\frac{v'+m'}{2}} \gamma^{\frac{v'+m'}{2}-1}}{\Gamma(m')\Gamma(v')\alpha^{\frac{v'+m'}{2}}} K_{v'-m'}\left[2\left(\frac{m'}{\alpha}\right)^{1/2} \sqrt{\gamma}\right],$ $v' = Lv, m' = Lm, \alpha = \Omega/v$
Nakagami- m	$f(r) = \frac{2}{\Gamma(m)} \left(\frac{m}{\Omega}\right)^m r^{2m-1} e^{-\frac{mr^2}{\Omega}}, r \geq 0,$ $\Gamma(m) = \int_0^\infty x^{m-1} \exp(-x) dx,$ $\sigma^2 = \frac{\Omega}{2} \left(1 - \sqrt{1 - m^{-1}}\right), m = E^2[r^2]/\text{var}(r^2),$ $f(\zeta) = \frac{m^m \zeta^{m-1}}{\zeta^m \Gamma(m)} \exp\left(-\frac{m\zeta}{\zeta}\right), \zeta \geq 0,$ $f(\gamma) = \left(\frac{m}{\gamma}\right)^{Lm} \frac{\gamma^{Lm-1}}{\Gamma(Lm)} \exp\left(-\frac{m}{\gamma}\right), \gamma \geq 0.$
$\eta - \mu$	$f(r) = \frac{4\sqrt{\pi}\mu^{\mu+\frac{1}{2}} h^\mu r^{2\mu}}{\Gamma(\mu) H^{\mu-\frac{1}{2}} \Omega^{\mu+\frac{1}{2}}} \exp\left(-\frac{2\mu h r^2}{\Omega}\right) I_{\mu-\frac{1}{2}}\left(\frac{2\mu H r^2}{\Omega}\right),$ $f(\zeta) = \frac{2\sqrt{\pi}\mu^{\mu+\frac{1}{2}} h^\mu \zeta^{\mu-\frac{1}{2}}}{\Gamma(\mu) H^{\mu-\frac{1}{2}} \zeta^{\mu+\frac{1}{2}}} \exp\left(-\frac{2\mu \zeta h}{\zeta}\right) \times I_{\mu-\frac{1}{2}}\left(\frac{2\mu H \zeta}{\zeta}\right),$ $f(\gamma) = \frac{2\sqrt{\pi} h^{L\mu}}{\Gamma(L\mu)} \left[\frac{L\mu}{\gamma}\right]^{L\mu-\frac{1}{2}} \exp\left(-\frac{2L\mu h \gamma}{\gamma}\right) \times I_{L\mu-\frac{1}{2}}\left(\frac{2L\mu H \gamma}{\gamma}\right)$
$\kappa - \mu$	$f(r) = \frac{2\mu(1+\kappa)^{\frac{\mu+1}{2}} r^\mu}{\kappa^{\frac{\mu-1}{2}} \exp(\mu\kappa) \Omega^{\frac{\mu+1}{2}}} \exp\left[-\frac{\mu(1+\kappa)r^2}{\Omega}\right] I_{\mu-1}\left[2\mu\sqrt{\frac{\kappa(1+\kappa)}{\Omega}} r\right],$ $f(\zeta) = \frac{\mu(1+\kappa)^{\frac{\mu+1}{2}} \zeta^{\frac{\mu-1}{2}}}{\kappa^{\frac{\mu-1}{2}} \exp(\mu\kappa) \zeta^{\frac{\mu+1}{2}}} \exp\left(-\frac{\mu(1+\kappa)\zeta}{\zeta}\right) \times I_{\mu-1}\left(2\mu\sqrt{\frac{\kappa(1+\kappa)}{\zeta}}\right),$ $f(\gamma) = L\mu \left[\frac{1+\kappa}{\gamma}\right]^{\frac{L\mu+1}{2}} \left[\frac{\gamma}{\kappa}\right]^{\frac{L\mu-1}{2}} \exp\left(-L\mu\left(\frac{(1+\kappa)\gamma}{\gamma} + \kappa\right)\right) \times I_{L\mu-1}\left(2L\mu\sqrt{\kappa(1+\kappa)}\frac{\gamma}{\gamma}\right)$

In these tables, we introduce the multipath fading channel and derive the expressions of typical performance metrics using the Meijer'G function as the unified form. The Meijer's G provides an alternative method in terms of the function elementary function and finite-range integrals of elementary functions. The Meijer'G function is represented as $G_{p,q}^{m,m}(z)$ and $G_{p,q}^{m,m}\left(z \left| \begin{matrix} a_1 & \dots & a_n \\ b_1 & \dots & b_q \end{matrix} \right. \right)$. In the closed form of BER and outage probability for different fading channels as in Tables 2 and 3, L denotes the number of branches. For the Rice fading channel, the closed-form of BER and outage probability are given in the infinite series form and converge rapidly.

In this paper, we focus on the typical fadings such as Rayleigh fading, Rice fading, K-fading, Nakagami- m fading, $\eta - \mu$ and $\kappa - \mu$ fading and derive unified formulas from Table 2 to Table 3. As we introduce in Section 3.2, the Hoyt fading, composite fading and

FTR fading are not included as well as the impulsive noise cases, which can be investigated in the future.

Table 2. BER for fading channels.

Channel Fading	BER
Rayleigh	$\frac{1}{2\sqrt{\pi}\Gamma(L)} G_{2,2}^{2,1} \left(\bar{\gamma} \left \begin{matrix} 1-L & 1 \\ 0 & \frac{1}{2} \end{matrix} \right. \right)$
Rice	$\frac{1}{2\sqrt{\pi}} e^{-\frac{s^2}{2\sigma^2}} \sum_{i=0}^{\infty} \frac{s^{2i}}{i!\Gamma(L+i)(2\sigma^2)^{L+2i}} \times G_{2,2}^{1,2} \left(\frac{1}{2\sigma^2} \left \begin{matrix} 1-l-i & \frac{1}{2}-l-i \\ 0 & 1-l-i \end{matrix} \right. \right)$
K-fading	$\frac{1}{2\sqrt{\pi}\Gamma(m)\Gamma(v)} G_{2,3}^{2,2} \left(\frac{m}{\alpha} \left \begin{matrix} 1 & \frac{1}{2} \\ v & m & 0 \end{matrix} \right. \right)$
Nakagami- m	$\frac{1}{2\sqrt{\pi}\Gamma(Lm)} \left(\frac{m}{\bar{\gamma}} \right)^{Lm} G_{2,2}^{2,1} \left(\frac{m}{\bar{\gamma}} \left \begin{matrix} 1-Lm & \frac{1}{2}-Lm \\ 0 & -Lm \end{matrix} \right. \right)$
$\eta - \mu$	$\frac{1}{\Gamma(L\mu)h^{L\mu}} \sum_{n=0}^{\infty} \frac{2^{-2n-2L\mu} H^{2n}}{n!h^{2n}\Gamma(n+L\mu+\frac{1}{2})} \times G_{2,2}^{1,2} \left(\frac{2L\mu h}{\bar{\gamma}} \left \begin{matrix} \frac{1}{2} & 1 \\ 2n+2L\mu & 0 \end{matrix} \right. \right)$
$\kappa - \mu$	$\frac{1}{2\sqrt{\pi} \exp(L\mu\kappa)} \sum_{n=0}^{\infty} \frac{(L\mu\kappa)^n}{n!\Gamma(n+L\mu)} \times G_{2,2}^{1,2} \left(\frac{L\mu(1+\kappa)}{\bar{\gamma}} \left \begin{matrix} \frac{1}{2} & 1 \\ n+L\mu & 0 \end{matrix} \right. \right)$

Table 3. Outage probability for fading channels.

Channel Fading	Outage Probability
Rayleigh	$\frac{\gamma_{th}^L}{\bar{\gamma}^L \Gamma(L)} G_{1,2}^{1,1} \left(\frac{1}{\bar{\gamma}} \left \begin{matrix} 1 & 0 & 0 \\ v & m & 0 \end{matrix} \right. \right)$
Rice	$e^{-\frac{s^2}{2\sigma^2}} \sum_{i=0}^{\infty} \frac{s^{2i}}{i!\Gamma(L+i)(2\sigma^2)^{L+2i}} \gamma_{th}^{L+i} \times G_{1,2}^{1,1} \left(\frac{\gamma_{th}}{2\sigma^2} \left \begin{matrix} 1-L-i & \\ 0 & -L-i \end{matrix} \right. \right)$
K-fading	$\frac{1}{\Gamma(m)\Gamma(v)} G_{1,3}^{2,1} \left(\frac{m}{\alpha} \gamma_{th} \left \begin{matrix} 1 & 0 & 0 \\ v & m & 0 \end{matrix} \right. \right)$
Nakagami- m	$\frac{(m\gamma_{th})^{Lm}}{\bar{\gamma}^{Lm}\Gamma(Lm)} G_{1,2}^{1,1} \left(\frac{m}{\bar{\gamma}} \gamma_{th} \left \begin{matrix} 1-Lm & \\ 0 & -Lm \end{matrix} \right. \right)$
$\eta - \mu$	$\frac{1}{\Gamma(m)\Gamma(v)} G_{1,3}^{2,1} \left(\frac{m}{\alpha} \gamma_{th} \left \begin{matrix} 1 & 0 & 0 \\ v & m & 0 \end{matrix} \right. \right)$
$\kappa - \mu$	$\frac{(m\gamma_{th})^{Lm}}{\bar{\gamma}^{Lm}\Gamma(Lm)} G_{1,2}^{1,1} \left(\frac{m}{\bar{\gamma}} \gamma_{th} \left \begin{matrix} 1-Lm & \\ 0 & -Lm \end{matrix} \right. \right)$

4.3. MRC Performance over Different Fading Channel

Figure 5 shows the simulated and theoretical BER comparisons versus average SNR with $L = 1, 2, 3$ receiver antennas over Rayleigh, Rice fading channel with Ricean factor $K = 5$ dB, K-fading channel with parameters $k = 1$, $v = 1$, respectively. It can be seen that for each kind of fading channel, increasing the order of diversity L improves the system performance. Due to the Rice model describing a signal with a dominant signal which means relatively better channel condition, the Rice fading channel obtains better BER compared to the K fading channel.

The outage probability over the Rice fading channel with Ricean factor K range from 0 to 10 dB and $L = 3$ branches for the MRC receiver is plotted in Figure 6. The SNR threshold is set to 0 dB. When the K factor equals -40 dB, which is equivalent to Rayleigh fading, the analytical and numerical results agree with Rayleigh's theoretical value well. It means the outage probability of Rayleigh distribution upperbounds that of the Rice fading channel. As the K factor increases, the increased power of the dominant received signal

decreases the outage probability, consequently increasing the diversity gain. In the figures, the analytical results show the correctness of derivation, and the results perfectly match.

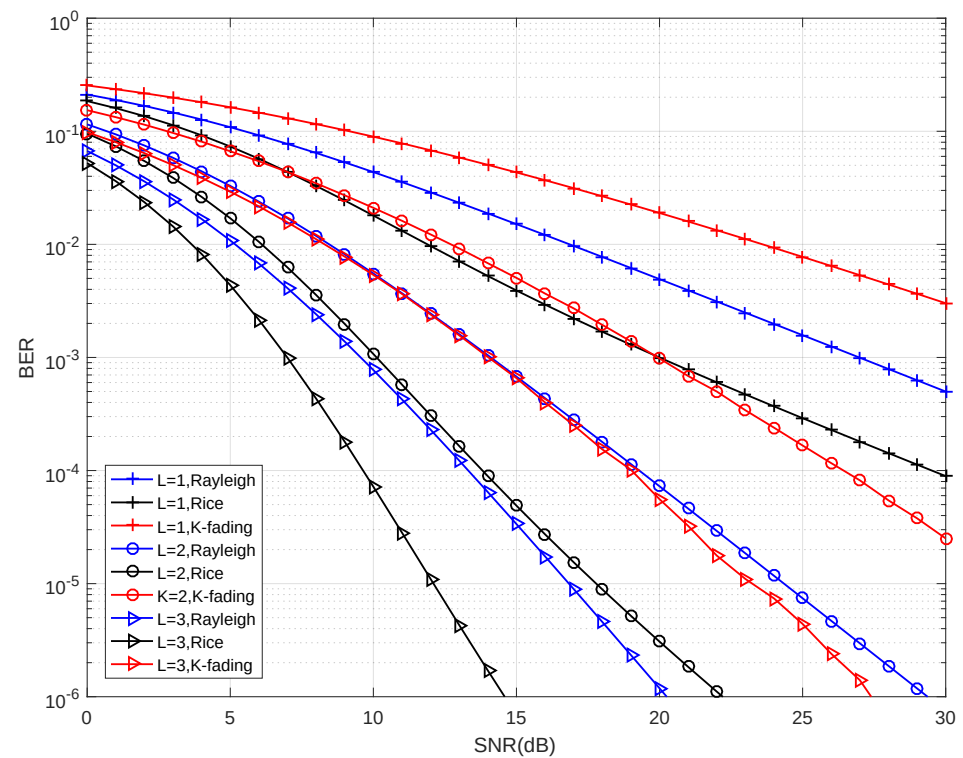


Figure 5. BER of MRC receiver over Rayleigh, Rice and K-fading channel.

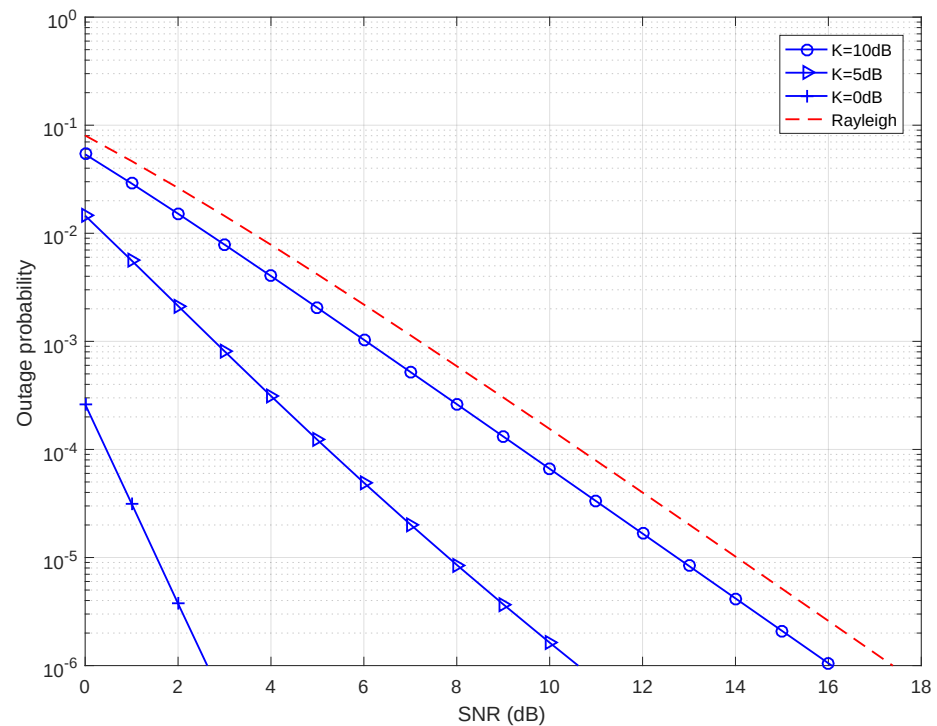


Figure 6. Outage probability of MRC receiver over Rice fading channel.

Numerical simulations are carried out to validate the theoretical analysis and simulated results over the Nakagami- m fading channel. Figure 7 shows BER results of MRC over the Nakagami- m fading channel with various parameters $m = 1, 1.5, 2$ and $L = 2$ branches. It

clearly shows the BER is consistent with that of the Rayleigh fading channel when $m = 1$, where m is the Nakagami fading parameter.

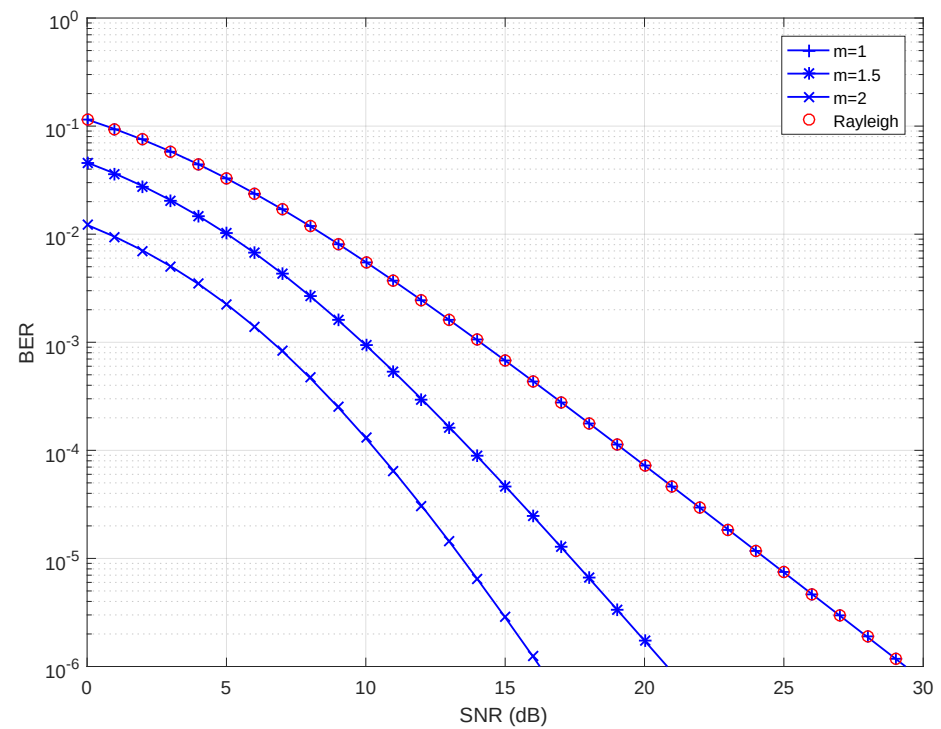


Figure 7. BER of MRC receiver for Nakagami- m fading channel.

The BER performances over $\eta - \mu$ fading channel with different η are shown in Figure 8. In the experiment, $L = 2$ and $\mu = 0.75$. When $\eta = 1$, the performance is approximated with that of the Nakagami- m fading channel, which is matched with the analysis before.

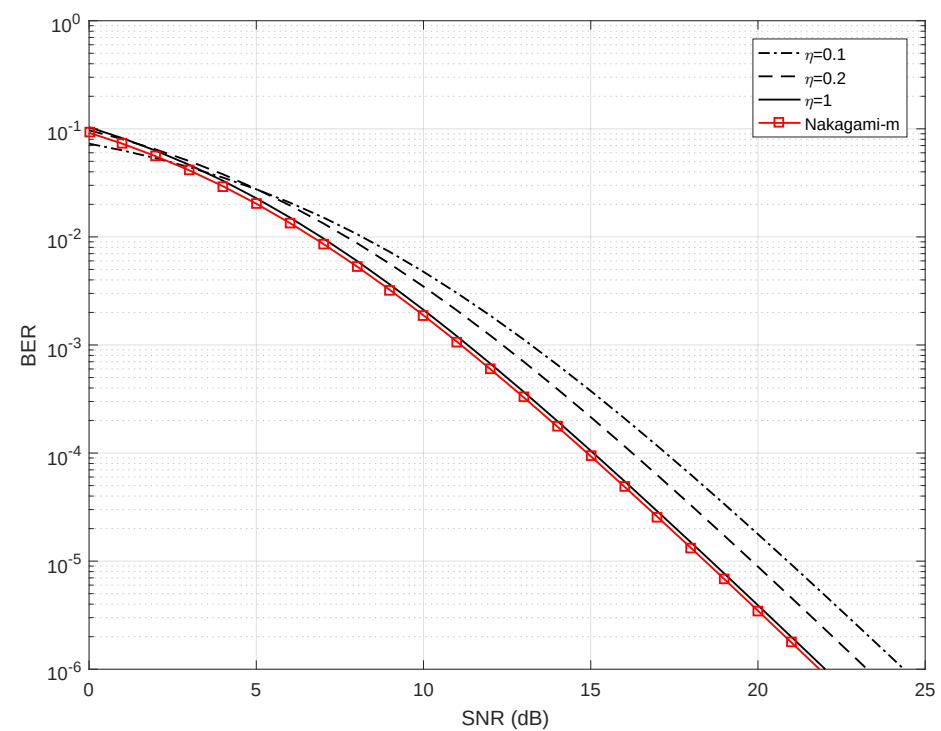


Figure 8. BER of MRC receiver over $\eta - \mu$ fading channel with $\eta = 0.1, 0.2, 1$.

K -distributed fading channel has a heavy tail for high amplitude distribution and more concentrated low amplitudes, which causes higher BER compared with Rayleigh fading. Figure 9 shows the BER performance over K -distributed fading channel with different shape parameters ν in the range [1 2.5 4]. When $\nu \rightarrow \infty$, K distribution becomes Rayleigh distribution.

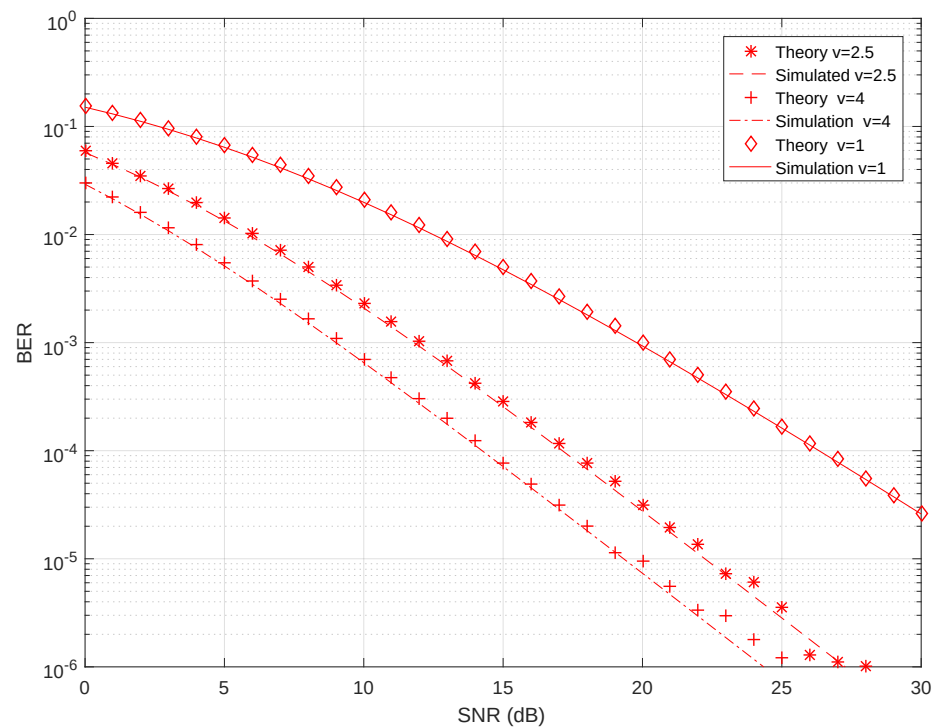


Figure 9. BER of MRC receiver over K -distributed fading channels with different fading parameter.

4.4. MRC Performance with CCI or Impulsive Noise

In complex scenarios, fading statistics for signal and interfering signals are more complicated due to different channel geometry as well as interference characteristics. Considering the case that the receiver obtains source signal and interfering signal simultaneously, the paper [91] studied MRC under CCI with arbitrary average power and the signal of interest (SOI) experienced arbitrary distribution in the presence of both CCI and AWGN. The works [92,93] studied the outage probability of MRC over the Rayleigh fading channel with CCI. The work [94] focused on the Rice fading channel with CCI. The work [95] analyzed the effects of CCI using MRC for the Nakagami fading channel. The work [96] considered independent CCIs with arbitrary powers and fading parameters as well as AWGN. The work [97] studied CCI effects on the digital radio systems in Nakagami fading channels in the presence of multiple equal-power CCIs and additive Gaussian noise. The paper [93] worked on the unequal-power CCI in correlated Rayleigh fading channel, and derived the outage probability of MRC under different correlation matrices. The work [98] derived the outage probability subject to $\eta - \mu$ fading, with the interfering signals having unequal power. The works [99,100] studied the outage probability and ABER of CCI-limited system in the Nakagami fading channel with the assumption that AWGN can be ignored. The work [101] considered the correlated $\eta - \mu$ fading channel and independent Rayleigh-fading CCI with arbitrary power in the presence of the background AWGN. The paper [102] tackled the performance analysis in arbitrarily correlated fading channels in the presence of multiple Rayleigh interferers. The paper [103] considered the MRC performance over correlated Nakagami- m fading channel for each branch in the presence of independent multiple interferers. The works [99,104] considered the performance of MRC diversity systems with Nakagami fading

in the presence of CCI. The work [98] derived the outage probability employing MRC with interfering signal subjected to $\eta - \mu$ distribution.

Different noise model has different parameters, the parameters of impulsive noise are generally unavailable and difficult to assume and estimate. The work [105] provided the closed-form expression of BER of linear diversity combining in the presence of impulsive noise. The works [106,107] gave a general analysis framework, which still cannot be applied to the S α S noise cases due to the requirement of known noise moments. The infinite variance causes the results to fail. The performance involved in the impulsive noise is analytically intractable, especially considering the channel fading simultaneously. The work [108] provided a unified analysis of diversity reception over composite fading channels and impulsive noise. The work [106] provided the BER performance analysis with non-Gaussian noise and interference.

Due to the lack of the optimality of linear combining schemes in the presence of impulsive noise, the locally optimal detectors in the presence of α noise are investigated [109]. The performance analysis will be much more complex and new mathematical methods are desired. Asymptotic complementary cumulative distribution function (CCDF) helps to solve the intractability of BER expressions. The work [110] provided the method based on the CCDF of stable distribution to obtain BER expression and provided the BER expressions when no closed-form expressions exist.

The work [111] derived the decision statistics of diversity in fading channels with S α S noise. The maximum possible diversity order of the scheme has relations with the characteristic exponent parameter α . The performance of combiners in the presence of S α S noise has been studied for multi-antenna and Rake receivers without analytical BER [112]. The work [113] investigated the LDPC-coded performance of diversity combining over non-Gaussian channels and derived exact BER of linear diversity combining on Rayleigh fading channels with impulsive noise.

5. Future Directions and Open Issues

Recently, many new concepts and communication systems have been developed such as vehicle networks, Internet of Things (IoT), massive MIMO, and low-power wide-area networks (LPWANs). Mmwave communications and multi-antenna technologies are very important and have been developed and applied in emerging fields to meet the requirements of communication reliability and latency. With the evolution of smart antennae and large-scale MIMO networks, MRC will still be a key element in constructing large systems, and there are many challenges and debates over the specifications of diversity combining in the future along with emerging techniques. Future works focus on more powerful approaches, such as the combinations with new disciplines (artificial intelligence techniques) and integration with new communication networks. Channel modeling, impulsive noise mitigation, or CCI suppression will find new ways and keep modifying. In this section, we will extend the future prospects from the traditional techniques to the emerging ones.

5.1. TAS/MRC

MIMO systems improve the spectral efficiency and robustness of communication systems. Without the degradation of full diversity, transmit antenna selection (TAS) can reduce the complexity. TAS combined with MRC forms TAS/MRC, which has been proposed as an efficient and low-complexity technique to modify the MIMO system [114]. TAS/MRC can achieve full transmit and receive diversity with only one radio frequency (RF) chain. At the receiver, the signals are processed following the MRC rule. Due to the requirement of one RF chain, TAS avoids the synchronization among antennas. Different transmit selection methods are investigated, i.e., the SNR-based selection which maximizes the MRC post-processed SNR at the receiver and the SINR-based selection which maximizes the post-processed SINR. The work [115] proposed an optimal TAS strategy to maximize the received signal-to-interference-plus-noise ratio (SINR) after MRC.

With perfectly known CSI between the primary user and secondary user, the closed form of performance of TAS/MRC is achieved. The work [116] analyzed the performance of TAS/MRC in traditional point-to-point communication systems and proved the maximal diversity order equals to the product of the number of transmit and receive antennas. With imperfect CSI, TAS is applied on the secondary transmitter and MRC is applied on the secondary receiver. The effective method has been adopted in massive MIMO systems to improve the robustness of massive MIMO systems [117]. The work [118] proposed hybrid access schemes that allow for sharing the same downlink resources between the small-cell network and the original macro-cell network. The author analyzed the achieved performance using the TAS/MRC in the presence of Poisson field interference. This algorithm covered the cases of highly correlated and un-correlated aggregated interference on different transmit channels. Also, the effect of delayed feedback and arbitrary TAS processing are investigated.

5.2. Imperfect CSI and Time-Varying Channel

In addition to channel characteristics inherently, CSI acquisition is always a key step in communication systems. Once the perfect CSI is achieved, the full diversity gain will be obtained for multi-channel systems. In the performance analysis of Section 2 or some derivations for the MRC diversity system in many works, one important assumption is the availability of perfect CSI at the receiver end. However, CSI errors cannot be avoided and CSI uncertainty exists in practical application for diversity systems. CSI errors mainly result from estimation methods and the time-varying nature of channels, which cause the failure of in-time and accurate CSI estimation as well as CSI feedback.

The ideal target of channel estimation is to achieve the perfect CSI and obtain exact information for signal decoding in a coherent modulation system. The work [119] investigated the impact of Gaussian estimation errors on the MRC performance considering multiple transmitters and receivers simultaneously. The pilot-aided channel estimation is one of the most famous methods, which has the advantages of improved spectral efficiency and adapting to time-varying channels [120]. The joint channel and data estimation scheme is promising for multiuser MIMO systems to further improve the estimation accuracy [121]. Time-varying channels cause outdated CSI and the phenomenon of channel aging. The channel between the transmitter and mobile receivers is unknown, and the feedback loop of CSI is very complicated. The work [122] studied the performance of the MIMO MRC system with feedback delay and channel estimation error. When combined with advanced communication schemes, the CSI uncertainty limits the diversity order of multi-channel transmissions. The paper [123] investigated secure cooperative transmissions in dual-hop MIMO relay system using the TAS/MRC scheme, which evaluates the impacts of outdated CSI and derives the closed form for ergodic secrecy rate and secrecy outage probability. The paper [14] studied the BER performance of OFDM-IM with a based greedy detector. The CSI uncertainty keeps affecting multiantenna and multiuser communication systems. The joint optimization of CSI and beamforming or CSI and precoding matrix will play an important part in advanced multiuser MIMO systems [124]. How to decrease the system sensitivity to CSI uncertainty is still an open issue for further modifications.

5.3. Beamforming and Combining

Spatial diversity can be achieved through transmitter diversity. Combined with receiver diversity, the double-side MRC is commonly referred to as MIMO-MRC [125]. Singular value decomposition (SVD)-based transmission beamforming combined with bit-interleaved coded modulation (BICM) can achieve full spatial multiplexing and full diversity gains with high computation complexity.

Beamforming structures can be classified as digital beamforming, analog beamforming and hybrid beamforming. The digital beamforming architecture needs many transceiver chains and has complicated hardware complexity and power-consuming components. The beamforming design and architecture of radio-frequency MRC (RF-MRB) provide

alternative methods for low-cost and low-complexity hardware in practical applications. RF-MRB only requires one downconversion RF chain.

Analog beamforming architecture processes the data stream and utilizes vector modulators including the components of phase splitter and variable gain amplifiers (VGAs) at each branch and signals are multiplied by the complex weight and finally added. To maximize the output SNR of the receive beamformer, the beamformer design for MIMO channels is highly non-convex.

Hybrid beamforming can retain performance benefits such as interference reduction as well as a reduction in cost and implementation complexity. A robust hybrid scheme can be designed, which combines null space projection (NSP) in the analog beamforming domain and the digital loading method in the digital beamforming domain. This method further improves the received SINR compared with separate analog beamforming and digital beamforming methods. Hybrid beamforming has the complementary pattern of multiple antennas compared with the distorted patterns of single antennas and compensates for other distortions introduced by the body and placement effects. Hybrid beamforming needs fewer RF chains than antennas at a lower cost and power.

The joint incorporation of beamforming and combining has advantages in terms of array gains and flexible receiver structure. In addition to combined MRC and beamforming, transceiver beamforming and new combining strategies are explored to emulate MRC at the receiver. The correlations of fading channels have detrimental effects for MRC. Maximal-ratio eigencombining (MREC) is a method to achieve the tradeoff to balance the MRC performance and complexity of beamforming over unknown correlations of fading channels. MREC combines the principle of MRC and beamforming, by projecting the received signal to the dominant eigenvectors of the channel correlation matrix and combining utilizing the MRC concepts. MREC can achieve comparable performance with MRC and beamforming, respectively, as their special cases, and also with the simplification of implementation. This eigencombining method is applied on the multi-antenna receiver end and the transmitter end. Similarly, the Rake receiver fits this eigencombining involved method, or utilizes the beamforming weight vector for each multipath of the desired user, and is combined using MRC.

5.4. Deep Learning-Based Combining

Deep learning (DL) emerges along with the explosive data and exploration for all kinds of unsolved problems. Recently, it has been applied in the communication field, also for the MIMO system. As described above, diversity suffers from multiple factors. How to achieve optimal or near-optimal performance considering multiple influential factors is a promising direction. Conventionally, finding optimal complex weight or beamforming solutions often needs iterative tracking in dynamic scenarios or non-Gaussian noisy scenarios. Iterative algorithms often have a high computational load and cannot satisfy the requirement of real-time realizations. This subsection focuses on the DL solutions aiming at the problems mentioned above and involves representative DL-based methods.

DL-based diversity combining can combat the effects of non-Gaussian noise and interference, which is applied to cases with unknown noise models and suitable for adaptive design. Training a fully connected deep neural network (DNN) offline with the dataset pairs consisting of CSI and channel impaired received signal and taking the detected symbols as the output allows us to consider non-Gaussian and interference for different branches simultaneously. The work [126] developed a DNN-based data detection algorithm for a diversity system, which is disturbed by different non-Gaussian noise and interference. The DNN is trained with different noisy datasets, which helps to avoid knowing the noise distribution in advance and achieve better detection performance than MRC. Additionally, another representative framework is the Bayesian estimation. Bayesian framework can realize the learning of parameters and the adaption to the instantaneous noise environment.

A hybrid beamforming structure can mitigate the hardware cost and power consumption. Currently, hybrid beamforming still suffers from some challenges. Hybrid

beamforming has the limits of high-computational complexity and lacks the exploitation for spatial information. When it comes to the optimization problem, analog and digital combiners are sequentially optimized. Iterative processing generally leads to sub-optimal solutions. The joint optimization of a hybrid beamforming system is a non-convex problem, especially for large-scale MIMO systems. DL has extraordinary performance in handling non-convex problems. To incorporate DL into hybrid beamforming, DNN can capture the structural information of hybrid beamforming and lower the computation complexity. The work [127] proposed an intelligent hybrid beamforming design method based on the autoencoder (AE) neural network. By mapping the hybrid beamforming system to an AE neural network, the original non-convex optimization problem is converted to the neural network training process. The work [128] proposed a hybrid precoder and combiner framework based on a convolutional neural network (CNN), which includes an analog precoding neural network, digital precoding neural network, analog combining network and digital combining neural network. With a designed supervised learning strategy, the joint network improves the hybrid precoding performance and maximizes spectral efficiency.

CSI is necessary for beamforming or combining, especially for large-scale MIMO systems. In many DL-based communication systems, the lack of incorporation of CSI will limit the applicable range of trained networks. Data-driven DL-based methods solve some scenarios for unknown channel models and parameters. For these cases, other artificial intelligence technologies such as reinforcement learning and meta-learning are the alternatives. To adapt to the varied channel conditions, meta-learning would provide a powerful method to adapt to new scenarios. Generative-Adversarial-Network (GAN) based framework allows one to learn channel transition probability and build an unknown channel model, which can adapt to the time-varying channels and realize channel tracking in dynamic channel conditions without being trained from scratch. The work [129] proposed a GAN variant-based channel estimation method, where two DL networks are trained adversarially with the received signals as the conditional input to learn an adaptive loss function. Furthermore, the GAN model can also learn the mapping from the received signals to the real channels. Moreover, for the known channel model, the combined DL-based method with the integration of domain knowledge is still beneficial.

Interference limits channel capacity and DL-based methods can exploit the structures of interference and then eliminate them. Model-driven DL-based method requires the distribution of interference, which is difficult to fit. Data-driven DL-based algorithms can exploit the correlation features to suppress the interference. For independent interfering signals, an improved deep convolutional neural network (DCNN)-based maximum likelihood detection (MLD) framework exploits the correlation features to suppress the interference [130], where MLD is used to produce an initial detection result and DCNN improves the detection by exploiting the local correlation to suppress the interference. Furthermore, the improved DCNN is enhanced by devising the loss function through the cross-entropy of the detection, which can help to suppress the interfering signals and simultaneously force the residual interference to approach Gaussian distribution.

Signal suffers from channel fading, noise, RF impairments, CCI and so on, blocked processing steps are developed traditionally, which cannot achieve optimal performance for the whole system. Each module works independently and influences each other. Also, many assumptions are made for non-ideal conditions such as AWGN or RF is perfect. DL provides us with methods to integrate the system. The trend of integration of multiple blocks that can interact with real environments will be overwhelming. DL methods can simultaneously process channel estimation, the equalization method and combat the inherent interferences. As more and more modules are integrated and more factors are considered, end-to-end optimization of communication systems is developed. A DL-based intelligent receiver replaces the traditional information recovery processes, utilizes the received signal inherently from the RF impairments, channel fading, noise, interference and other factors in a dynamic environment and realizes blind reception. This concept is popular for SISO systems, and the coordination of multiple transmitters and receivers is still a problem to explore, such as which parts are integrated and which factors are considered. Also, the

extensions from SISO to MIMO, from single-subcarriers to multisubcarriers, and from MIMO to large-scale MIMO, are issues worthy of being solved.

5.5. Development with Other Systems

Signals are transmitted over parallel multi-channels and diversity gains are achieved through combining techniques. Due to the wide spread of 5G communications and on the road to 6G, the multi-antenna system will be integrated with new burgeoning systems. In the future, the system will extend to high-frequency communications, where the increased frequency degrades the ability of diffraction. The relaying technique is a powerful technique that increases the data rate and user quality and further extends the coverage of the signal. As a brick to build the complete communication system, diversity combining will help. This subsection provides technologies that include MRC and briefly lists the development directions in the future.

With the wide spread of cellular networks, multiuser MIMO systems and massive MIMO have attracted researchers' attention, where the base station is equipped with multiple antennas and sends signals to a large number of users. Multiple access is one key technology for multiuser communication networks to satisfy the requirement of high data rate and high spectral efficiency. As the scale of the system enlarges, the inter-user interference and noise will be worse. Similarly, with multi-antenna diversity, multiuser diversity can be utilized under the fact that the deep fading of all channels between the base station and users cannot occur simultaneously. The concatenation of multiuser-diversity and MIMO naturally enhances the performance of these two technologies. The paper [131] studied the multicarrier code division multiple access (MC-CDMA) system which combines the diversity of frequency, space and code domain. The MC-CDMA with antenna array presents good performance in signal reception and interference cancellation. Currently, Non-Orthogonal Multiple Access (NOMA) is a promising candidate for future generations of mobile networks with the guarantee of massive connectivity and low latency transmission. Combining NOMA with MIMO MRC or other mmWave MIMO techniques will provide new alternatives for high throughput and spectral efficiency.

As the IoTs develop, scarce frequency and power resources are required to be multiplexed among diverse sensors and millions of users are connected to networks covering large areas. LoRa is a promising technology for the wide-area IoTs [132], which allows multiple users accommodated in one channel via multiple access techniques. However, the performance of the LoRa network is constrained by multiple user interference and signal collisions. Also, with the requirements for data rate increasing, the tradeoff between coverage and throughput is difficult to achieve especially in the case of energy constraints over fading environments. Multi-antenna technologies or MIMO-LoRa concept provides alternative schemes for high data rate links and novel system architectures as well as link reliability. Consequently, signal combinations and detection for novel LoRa systems are necessary to design and analyze.

Cognitive radio (CR) is a promising technique to improve the spectrum efficiency of new-generation communication systems. CR can help to realize the spectrum resource management to effectively avoid interference between the secondary user (SU) and the primary user (PU). For a CR network, the CCIs are existing in links between the second user (SU) and the primary user (PU), from PU to SU, among SUs. With the increasing trend of multi-antenna applications, severe channel conditions such as correlations will be common factors for system design which have detrimental effects on the performance. As for the emerging multi-antenna CR wireless communication, the issue is much more obvious. To further relax the assumption of i.i.d fading channel among branches, emerging multi-antenna systems cannot ignore the correlation among subbranches, which will degrade the detector performance.

Chaotic communications are the representative techniques for short-range or secure communications with randomness, low power and low-cost characteristics. Conventional chaotic communication schemes have low energy efficiency and low data rates. Currently,

chaotic communication schemes are designed for ultra-wideband communication, MIMO, or OFDM scheme [133]. Besides the design of coded-modulation schemes in chaotic communications, the combination with multiuser MIMO, and index modulation aided multicarrier systems with the potential to utilize multiple kinds of transmission entities in time, frequency and space domains to improve spectral efficiency.

The energy harvesting technique is also an emerging solution to confront the energy dilemma for energy-constrained systems like wireless or underwater devices. The RF signals themselves can be seen as energy sources. The nature of the RF signal makes CCI become the potential energy source. Dual-hop AF relay systems equip multiple antennas for the relay node, and the source and destination nodes both have one antenna. Here, it can utilize CCI as additional energy. Exploiting CCI as a beneficial prospect for MRC development, the energy harvesting dual-hop relaying system is promising.

As wireless communications move toward the 6G era, sub-THz and THz communications have emerged as key candidates in future-generation communications. However, the sub-THz and THz-band signals are highly sensitive to blockages and the beams in diverse directions may generate pointing errors, which leads to wrong combining weights for detection. Reconfigurable intelligent surface (RIS) integrates tunable passive elements and can adjust the scattering properties of reflecting elements, which can promisingly control the directions of THz-signals and reduce path attenuation from high frequencies [134]. FTR fading model is necessary for the THz channel modeling and fitting for the measured channel data. Exploiting the RIS-aided scheme and finding the design concept and performance bounds over the FTR fading channel are required to be further explored.

6. Conclusions

In this survey article, we have focused on the MRC among linear combining techniques, and reviewed several aspects including major diversity models for applications. Based on the analysis model, the channel fading statistics and performance metrics, the performance analysis for representative channel models in terms of BER and outage probability are derived and closed forms are represented. Besides the channel fading statistics, we examine a range of influential factors such as channel correlation, CCI and impulsive noise. In the future, new methods are reviewed based on the issues existing in the system within the framework of traditional and emerging techniques. Traditional methods such as TAS/MRC or beamforming will continue find their place due to the advantages of low complexity and high diversity gain. DL methods improve performance and provide ways to combat dynamic or more complicated environments, and many issues will be solved by integrating system blocks. Furthermore, we highlight the integration of diversity combined with a new wireless network and new application scenarios. In the future, MRC analysis over FTR fading models and DL based design for MRC combined mmWave communication receiver will be included in our next-step work.

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